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(12) **United States Patent**
**Wada et al.**

(10) **Patent No.: US 12,399,730 B2**
(45) **Date of Patent: Aug. 26, 2025**

(54) **SCALING SYSTEM AND CALCULATION METHOD**

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(72) Inventors: **Kiyomi Wada**, Tokyo (JP); **Shinichi Hayashi**, Tokyo (JP)

(73) Assignee: **HITACHI VANTARA, LTD.**, Kanagawa-Ken (JP)

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(21) Appl. No.: **17/469,605**

(22) Filed: **Sep. 8, 2021**

(65) **Prior Publication Data**

US 2022/0156101 A1 May 19, 2022

(30) **Foreign Application Priority Data**

Nov. 13, 2020 (JP) ................................ 2020-189197

(51) **Int. Cl.**
***G06F 9/455*** (2018.01)
***G06F 9/355*** (2018.01)

(52) **U.S. Cl.**
CPC ........ ***G06F 9/45558*** (2013.01); ***G06F 9/3555*** (2013.01); *G06F 2009/4557* (2013.01); *G06F 2009/45575* (2013.01)

(58) **Field of Classification Search**
CPC ... G06F 2009/4557; G06F 2009/45575; G06F 9/5077
See application file for complete search history.

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*Primary Examiner* — Dong U Kim
(74) *Attorney, Agent, or Firm* — Volpe Koenig

(57) **ABSTRACT**

A scaling system ranges over a first base in which a first calculating apparatus having first and second worker nodes operates, and a second base in which a storage apparatus connected to the first base by a network is set. The storage apparatus includes first and second network ports, and first and second volumes accessed by first and second worker node, respectively. A second calculating apparatus is further set in the first base, and the second worker node is moved to the second calculating apparatus and operates as a third worker node and the second volume communicates with the third worker node through the second network port if the transfer rate of the first calculating apparatus or the transfer rate of the first network port exceeds a predetermined threshold when the first and second volumes are in communication with the first calculating apparatus through the first network port.

**10 Claims, 18 Drawing Sheets**

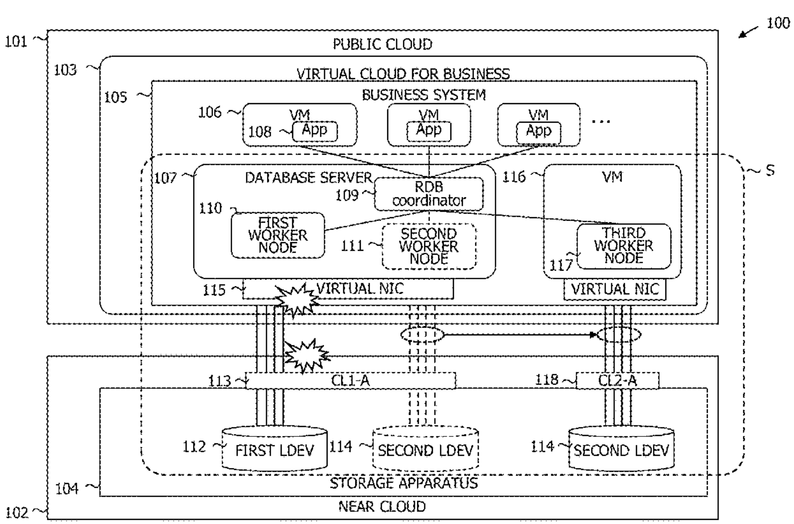

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FIG. 1

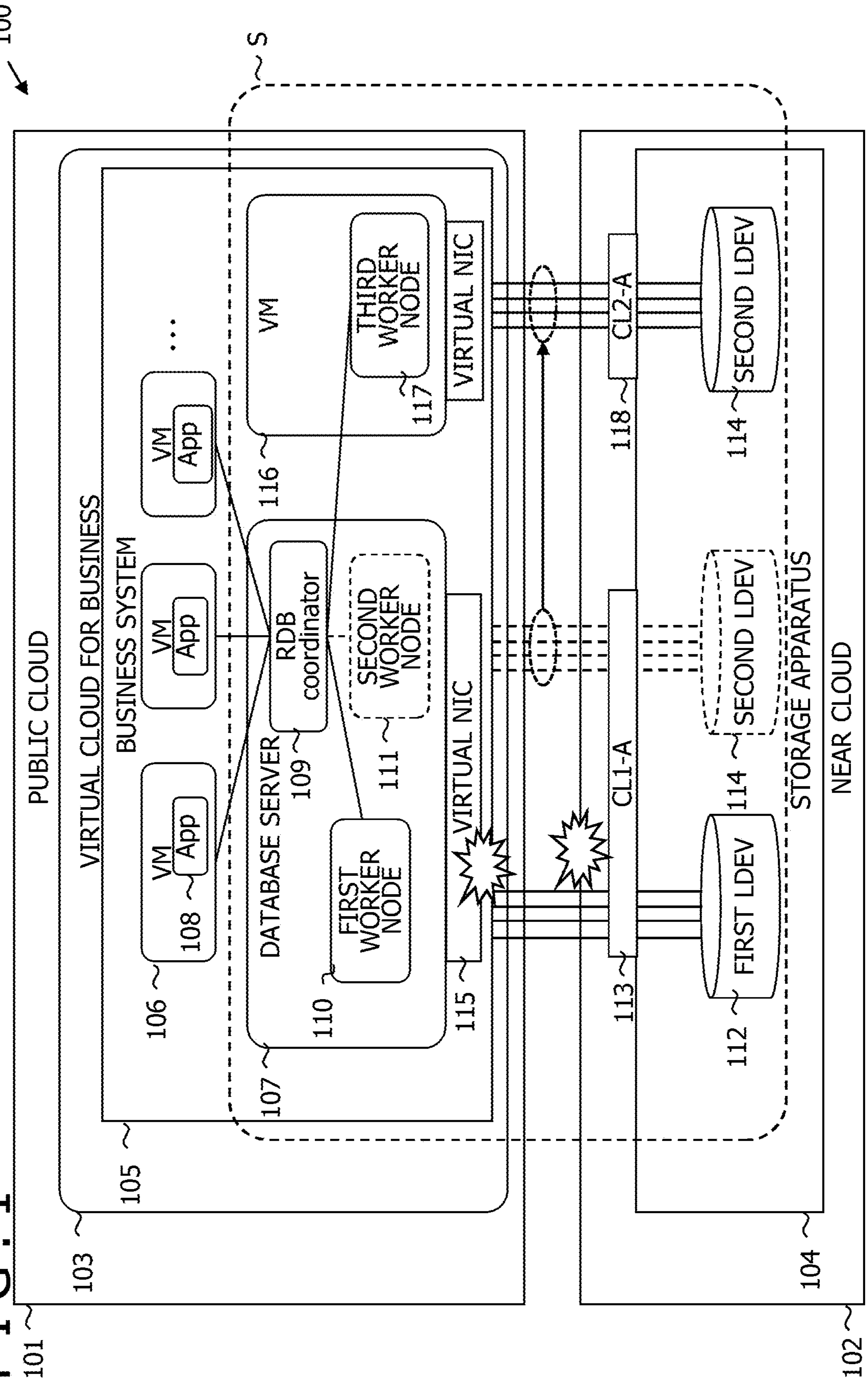
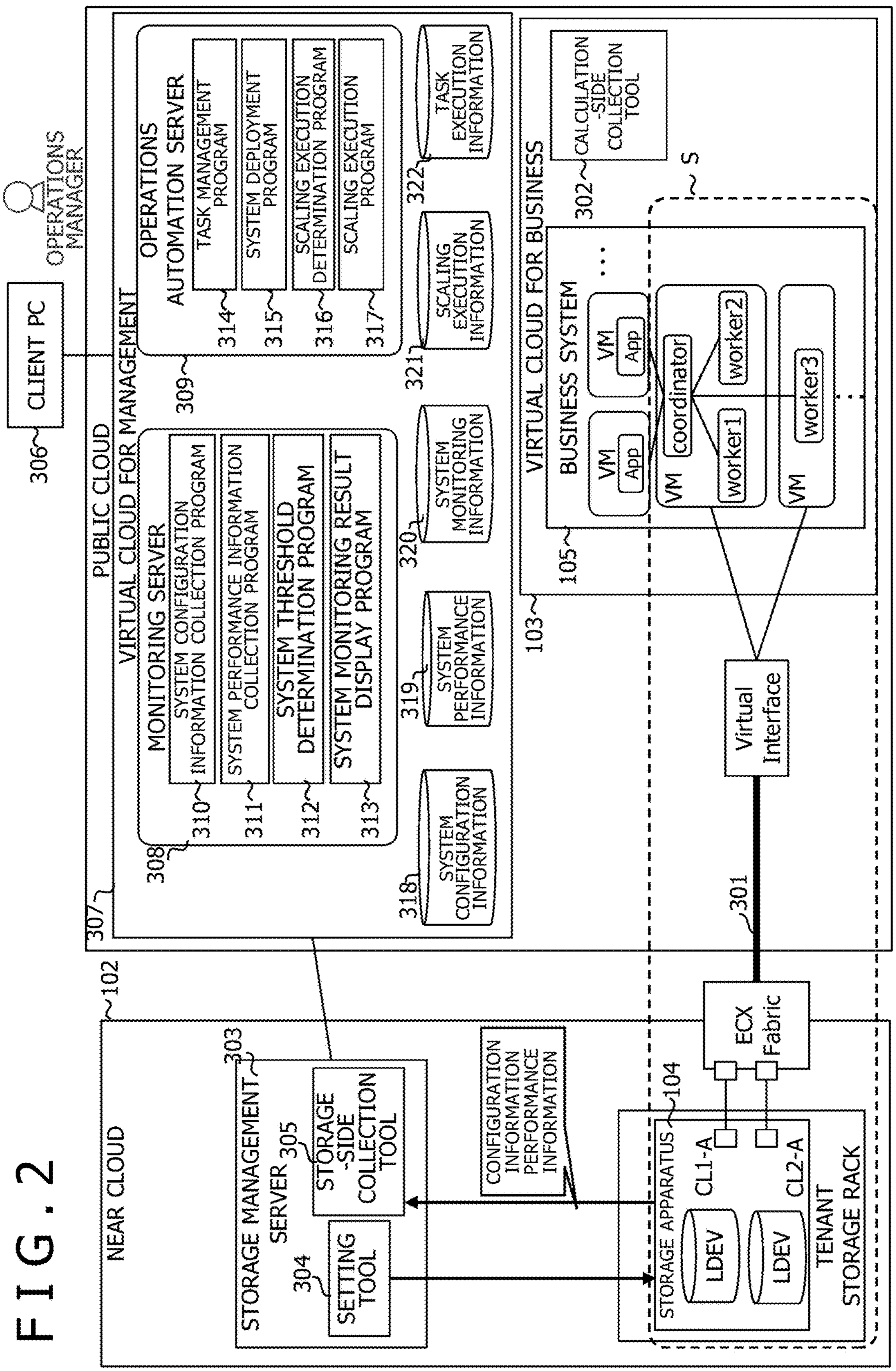

100
S
PUBLIC CLOUD
VIRTUAL CLOUD FOR BUSINESS
BUSINESS SYSTEM
VM
App
VM
App
VM
App
DATABASE SERVER
RDB coordinator
FIRST WORKER NODE
SECOND WORKER NODE
VM
THIRD WORKER NODE
VIRTUAL NIC
VIRTUAL NIC
CL1-A
CL2-A
FIRST LDEV
SECOND LDEV
SECOND LDEV
STORAGE APPARATUS
NEAR CLOUD
101
102
103
104
105
106
107
108
109
110
111
112
113
114
114
115
116
117
118

FIG. 2



OPERATIONS MANAGER
CLIENT PC
306
PUBLIC CLOUD
307
VIRTUAL CLOUD FOR MANAGEMENT
MONITORING SERVER
308
SYSTEM CONFIGURATION INFORMATION COLLECTION PROGRAM
310
SYSTEM PERFORMANCE INFORMATION COLLECTION PROGRAM
311
SYSTEM THRESHOLD DETERMINATION PROGRAM
312
SYSTEM MONITORING RESULT DISPLAY PROGRAM
313
OPERATIONS AUTOMATION SERVER
309
TASK MANAGEMENT PROGRAM
314
SYSTEM DEPLOYMENT PROGRAM
315
SCALING EXECUTION DETERMINATION PROGRAM
316
SCALING EXECUTION PROGRAM
317
SYSTEM CONFIGURATION INFORMATION
318
SYSTEM PERFORMANCE INFORMATION
319
SYSTEM MONITORING INFORMATION
320
SCALING EXECUTION INFORMATION
321
TASK EXECUTION INFORMATION
322
VIRTUAL CLOUD FOR BUSINESS
103
BUSINESS SYSTEM
105
VM
App
VM
App
VM
coordinator
worker1
worker2
VM
worker3
CALCULATION -SIDE COLLECTION TOOL
302
S
Virtual Interface
301
ECX Fabric
NEAR CLOUD
102
STORAGE MANAGEMENT SERVER
303
STORAGE -SIDE COLLECTION TOOL
305
SETTING TOOL
304
CONFIGURATION INFORMATION PERFORMANCE INFORMATION
TENANT STORAGE RACK
STORAGE APPARATUS
104
CL1-A
CL2-A
LDEV
LDEV

FIG. 3

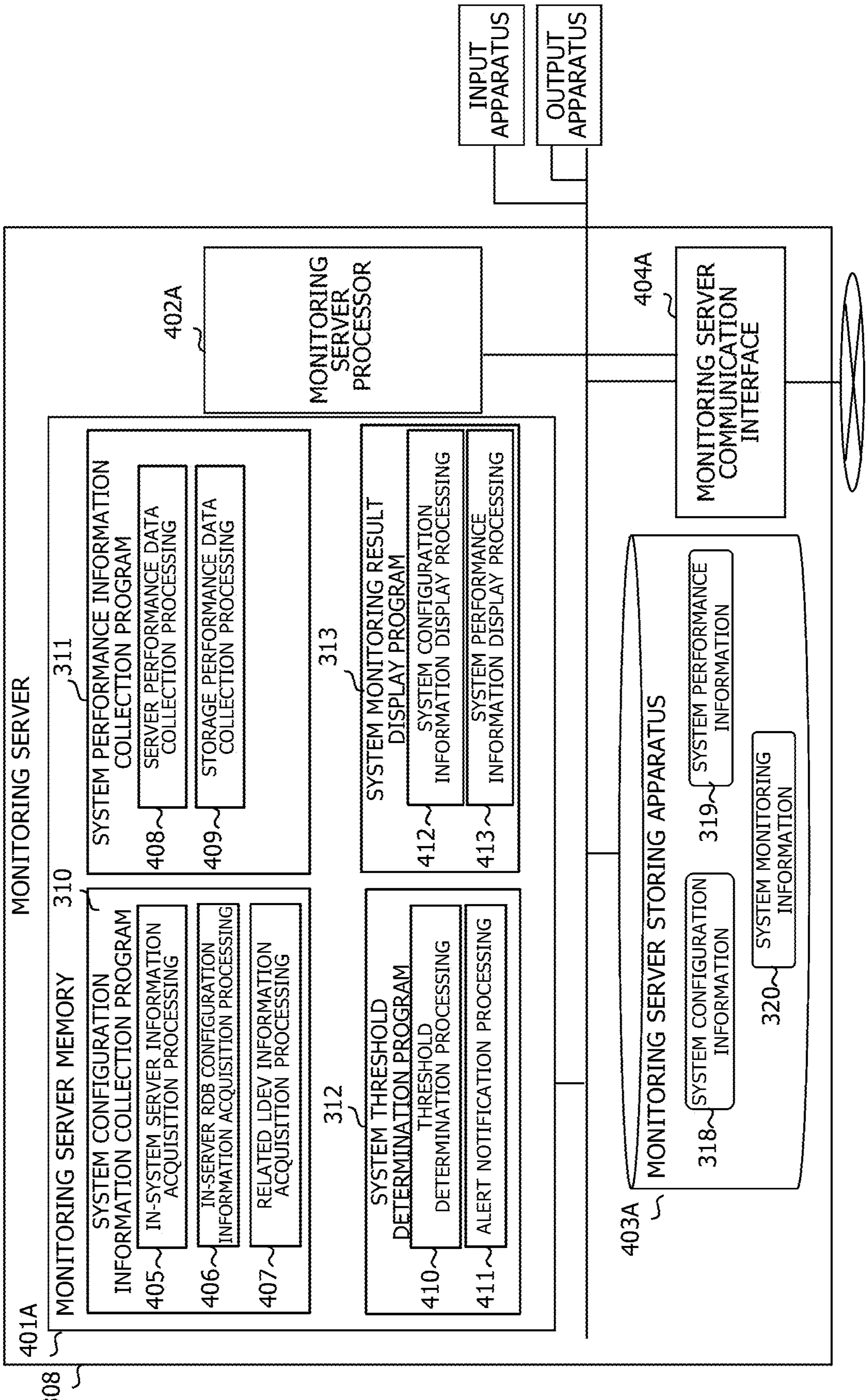

308
401A
MONITORING SERVER
MONITORING SERVER MEMORY
310
SYSTEM CONFIGURATION INFORMATION COLLECTION PROGRAM
405
IN-SYSTEM SERVER INFORMATION ACQUISITION PROCESSING
406
IN-SERVER RDB CONFIGURATION INFORMATION ACQUISITION PROCESSING
407
RELATED LDEV INFORMATION ACQUISITION PROCESSING
311
SYSTEM PERFORMANCE INFORMATION COLLECTION PROGRAM
408
SERVER PERFORMANCE DATA COLLECTION PROCESSING
409
STORAGE PERFORMANCE DATA COLLECTION PROCESSING
312
SYSTEM THRESHOLD DETERMINATION PROGRAM
410
THRESHOLD DETERMINATION PROCESSING
411
ALERT NOTIFICATION PROCESSING
313
SYSTEM MONITORING RESULT DISPLAY PROGRAM
412
SYSTEM CONFIGURATION INFORMATION DISPLAY PROCESSING
413
SYSTEM PERFORMANCE INFORMATION DISPLAY PROCESSING
402A
MONITORING SERVER PROCESSOR
404A
MONITORING SERVER COMMUNICATION INTERFACE
403A
MONITORING SERVER STORING APPARATUS
318
SYSTEM CONFIGURATION INFORMATION
319
SYSTEM PERFORMANCE INFORMATION
320
SYSTEM MONITORING INFORMATION
INPUT APPARATUS
OUTPUT APPARATUS

FIG. 4

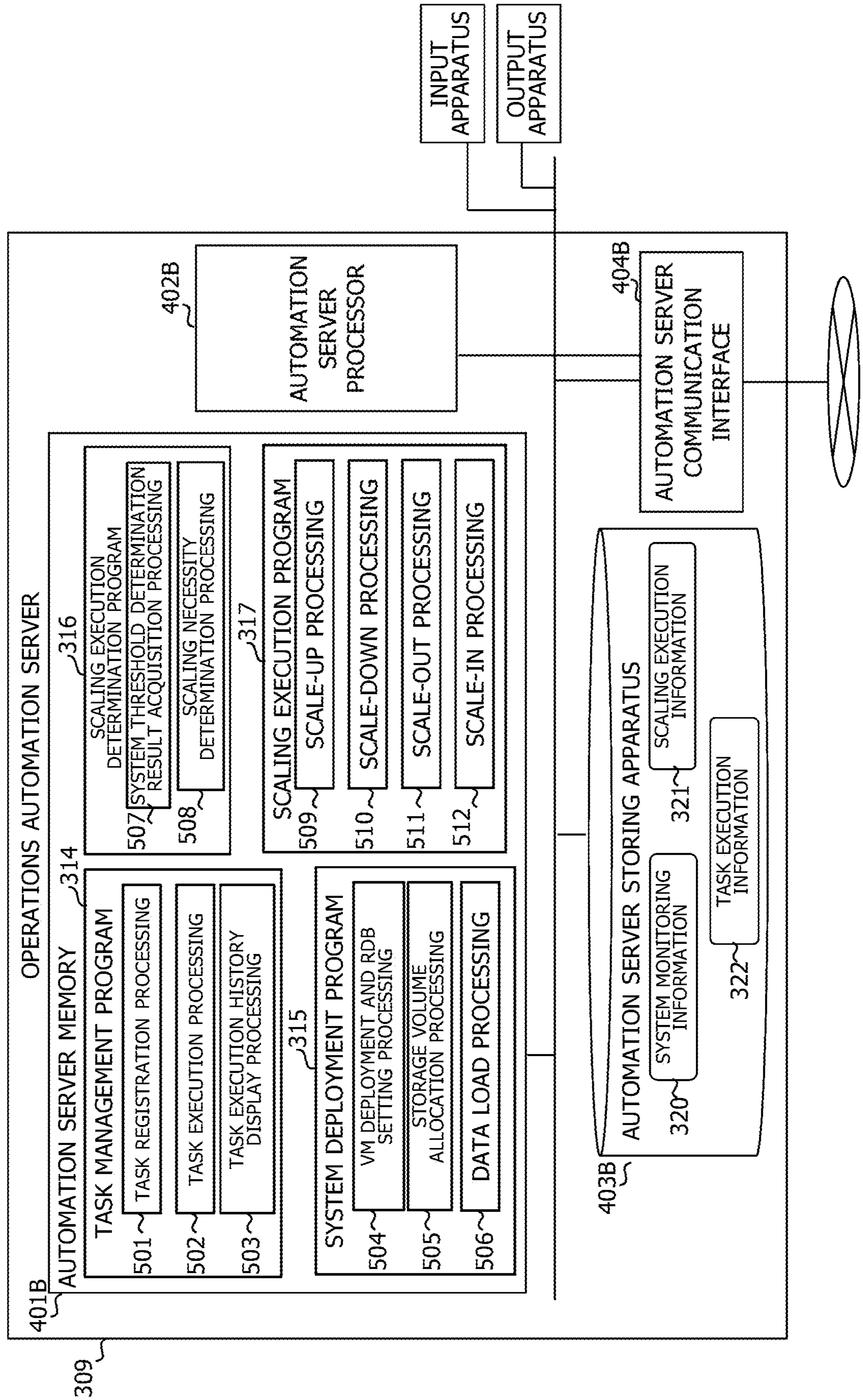
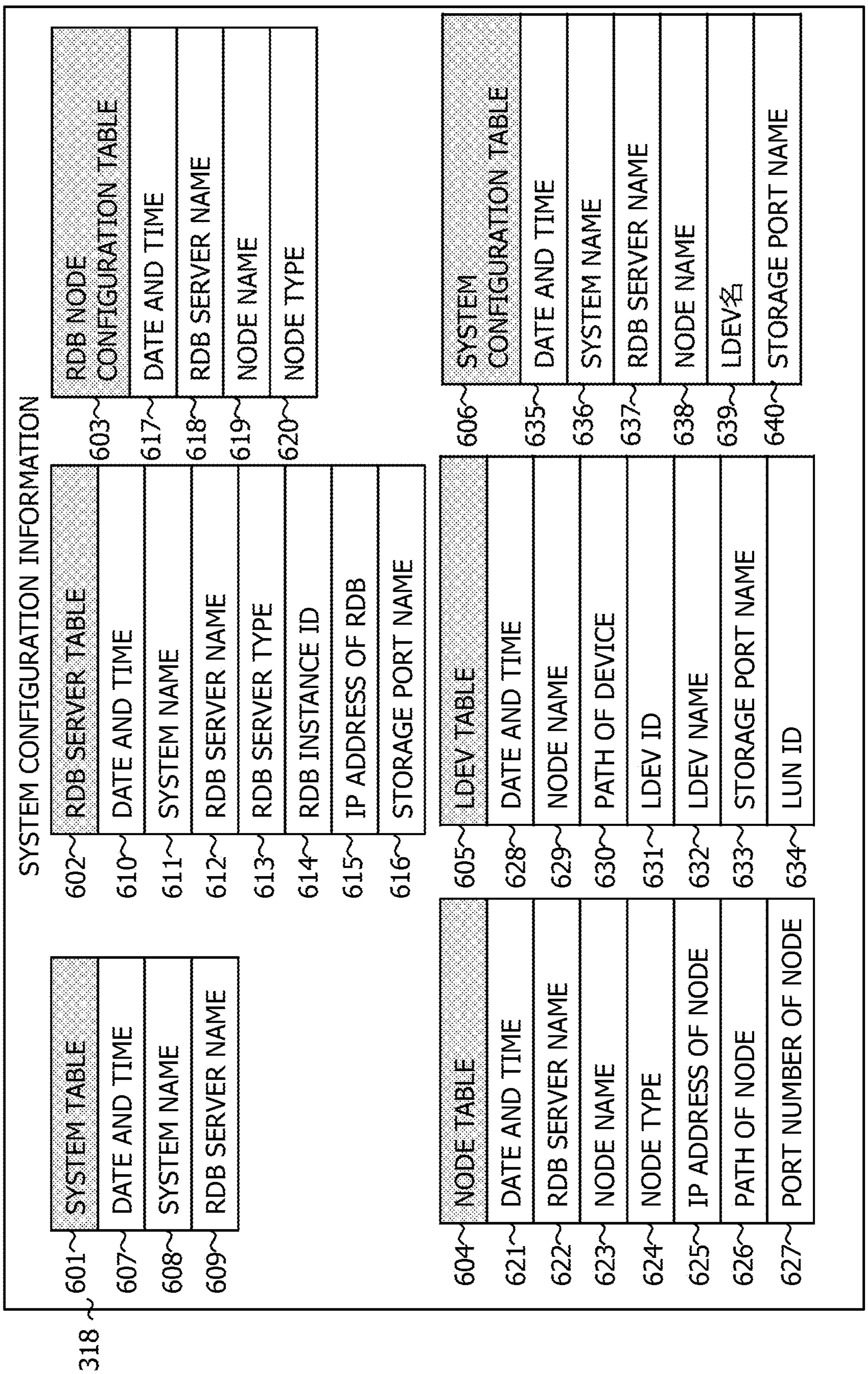
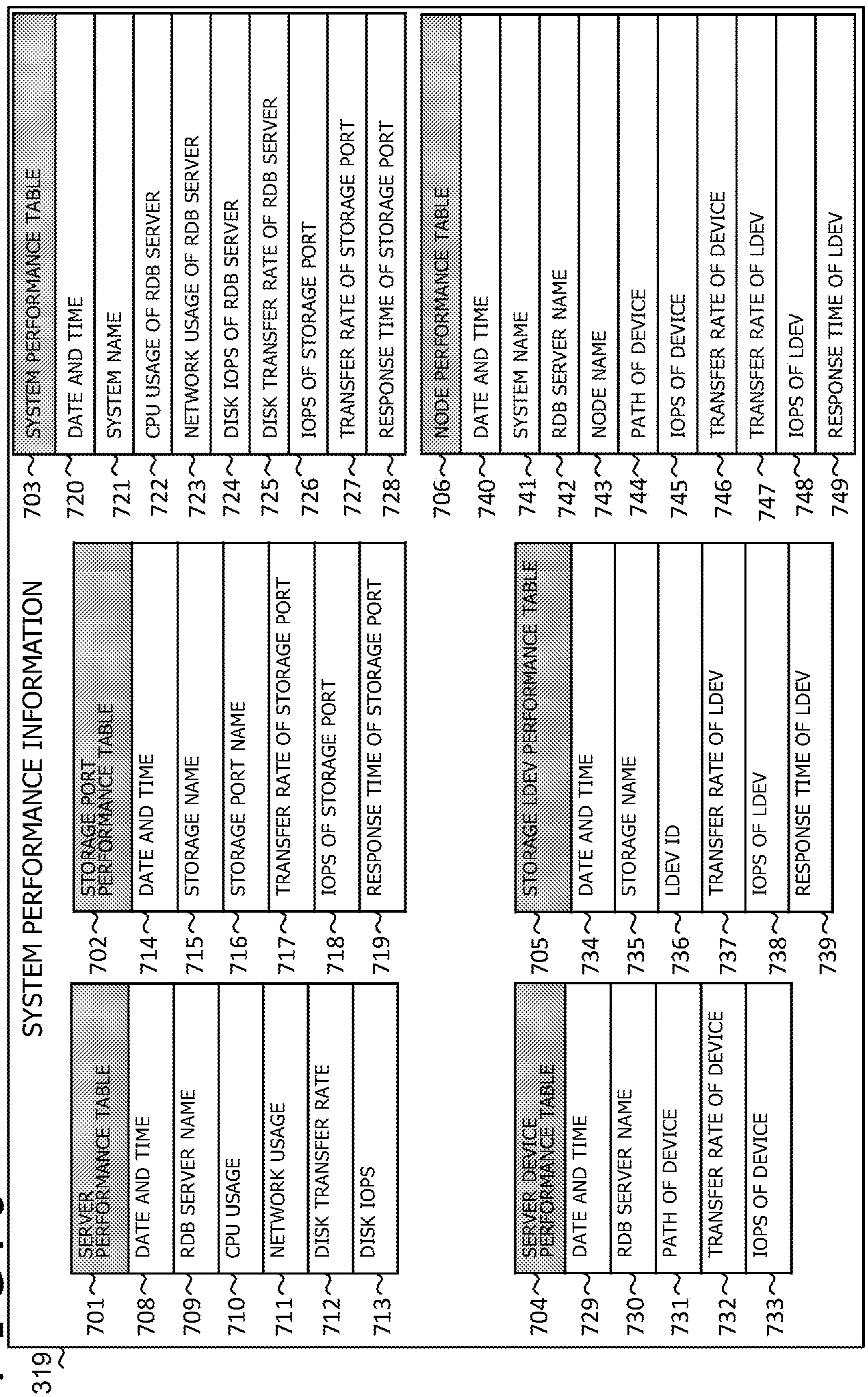

INPUT APPARATUS
OUTPUT APPARATUS
309
OPERATIONS AUTOMATION SERVER
401B
AUTOMATION SERVER MEMORY
314
TASK MANAGEMENT PROGRAM
501 TASK REGISTRATION PROCESSING
502 TASK EXECUTION PROCESSING
503 TASK EXECUTION HISTORY DISPLAY PROCESSING
315
SYSTEM DEPLOYMENT PROGRAM
504 VM DEPLOYMENT AND RDB SETTING PROCESSING
505 STORAGE VOLUME ALLOCATION PROCESSING
506 DATA LOAD PROCESSING
316
SCALING EXECUTION DETERMINATION PROGRAM
507 SYSTEM THRESHOLD DETERMINATION RESULT ACQUISITION PROCESSING
508 SCALING NECESSITY DETERMINATION PROCESSING
317
SCALING EXECUTION PROGRAM
509 SCALE-UP PROCESSING
510 SCALE-DOWN PROCESSING
511 SCALE-OUT PROCESSING
512 SCALE-IN PROCESSING
402B
AUTOMATION SERVER PROCESSOR
404B
AUTOMATION SERVER COMMUNICATION INTERFACE
403B
AUTOMATION SERVER STORING APPARATUS
320 SYSTEM MONITORING INFORMATION
321 SCALING EXECUTION INFORMATION
322 TASK EXECUTION INFORMATION

FIG. 5



318
SYSTEM CONFIGURATION INFORMATION
601 SYSTEM TABLE
607 DATE AND TIME
608 SYSTEM NAME
609 RDB SERVER NAME
602 RDB SERVER TABLE
610 DATE AND TIME
611 SYSTEM NAME
612 RDB SERVER NAME
613 RDB SERVER TYPE
614 RDB INSTANCE ID
615 IP ADDRESS OF RDB
616 STORAGE PORT NAME
603 RDB NODE CONFIGURATION TABLE
617 DATE AND TIME
618 RDB SERVER NAME
619 NODE NAME
620 NODE TYPE
604 NODE TABLE
621 DATE AND TIME
622 RDB SERVER NAME
623 NODE NAME
624 NODE TYPE
625 IP ADDRESS OF NODE
626 PATH OF NODE
627 PORT NUMBER OF NODE
605 LDEV TABLE
628 DATE AND TIME
629 NODE NAME
630 PATH OF DEVICE
631 LDEV ID
632 LDEV NAME
633 STORAGE PORT NAME
634 LUN ID
606 SYSTEM CONFIGURATION TABLE
635 DATE AND TIME
636 SYSTEM NAME
637 RDB SERVER NAME
638 NODE NAME
639 LDEV名
640 STORAGE PORT NAME

FIG. 6



319
SYSTEM PERFORMANCE INFORMATION
701 SERVER PERFORMANCE TABLE
708 DATE AND TIME
709 RDB SERVER NAME
710 CPU USAGE
711 NETWORK USAGE
712 DISK TRANSFER RATE
713 DISK IOPS
702 STORAGE PORT PERFORMANCE TABLE
714 DATE AND TIME
715 STORAGE NAME
716 STORAGE PORT NAME
717 TRANSFER RATE OF STORAGE PORT
718 IOPS OF STORAGE PORT
719 RESPONSE TIME OF STORAGE PORT
703 SYSTEM PERFORMANCE TABLE
720 DATE AND TIME
721 SYSTEM NAME
722 CPU USAGE OF RDB SERVER
723 NETWORK USAGE OF RDB SERVER
724 DISK IOPS OF RDB SERVER
725 DISK TRANSFER RATE OF RDB SERVER
726 IOPS OF STORAGE PORT
727 TRANSFER RATE OF STORAGE PORT
728 RESPONSE TIME OF STORAGE PORT
704 SERVER DEVICE PERFORMANCE TABLE
729 DATE AND TIME
730 RDB SERVER NAME
731 PATH OF DEVICE
732 TRANSFER RATE OF DEVICE
733 IOPS OF DEVICE
705 STORAGE LDEV PERFORMANCE TABLE
734 DATE AND TIME
735 STORAGE NAME
736 LDEV ID
737 TRANSFER RATE OF LDEV
738 IOPS OF LDEV
739 RESPONSE TIME OF LDEV
706 NODE PERFORMANCE TABLE
740 DATE AND TIME
741 SYSTEM NAME
742 RDB SERVER NAME
743 NODE NAME
744 PATH OF DEVICE
745 IOPS OF DEVICE
746 TRANSFER RATE OF DEVICE
747 TRANSFER RATE OF LDEV
748 IOPS OF LDEV
749 RESPONSE TIME OF LDEV

FIG. 7

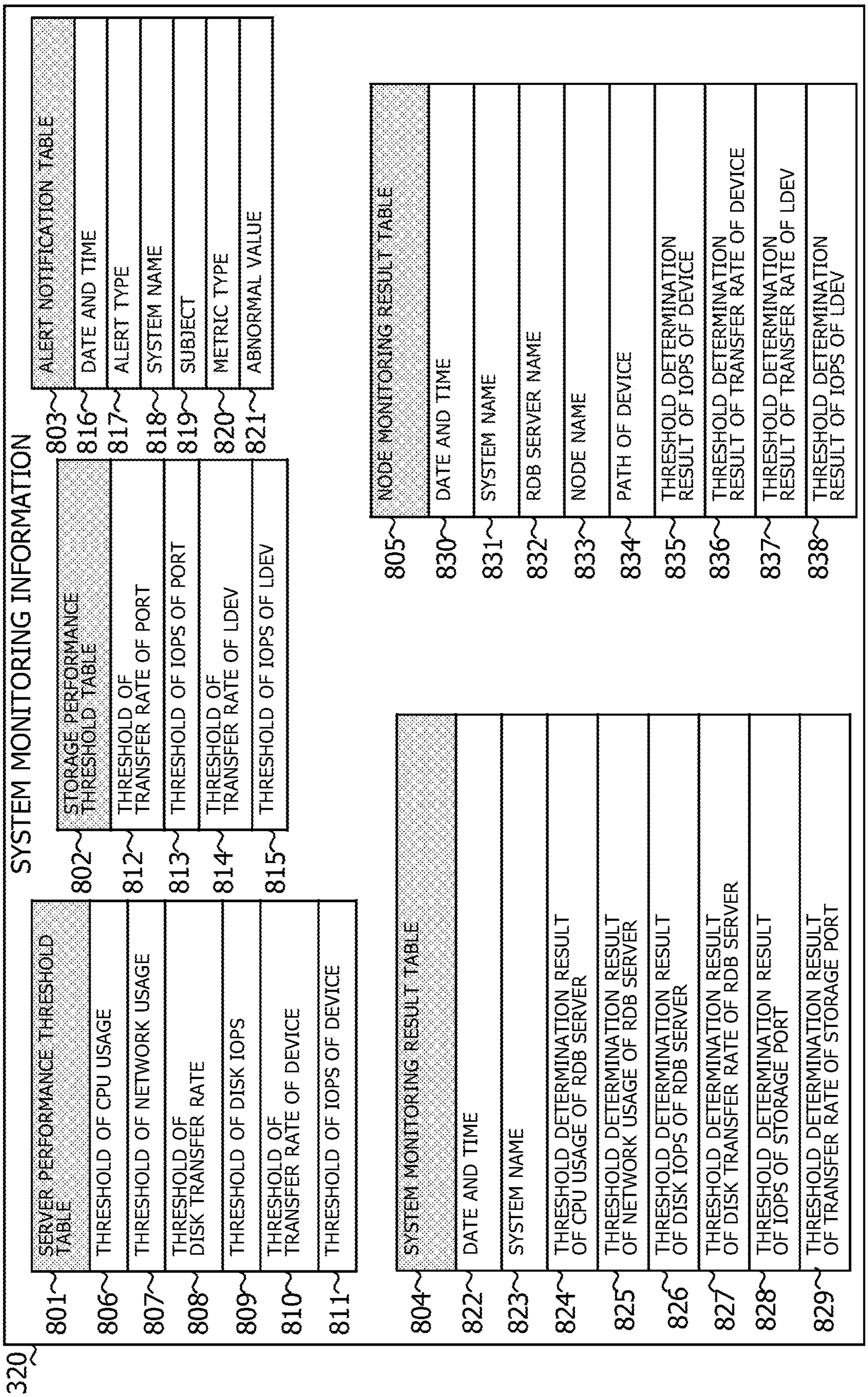

320
SYSTEM MONITORING INFORMATION
801 SERVER PERFORMANCE THRESHOLD TABLE
806 THRESHOLD OF CPU USAGE
807 THRESHOLD OF NETWORK USAGE
808 THRESHOLD OF DISK TRANSFER RATE
809 THRESHOLD OF DISK IOPS
810 THRESHOLD OF TRANSFER RATE OF DEVICE
811 THRESHOLD OF IOPS OF DEVICE
802 STORAGE PERFORMANCE THRESHOLD TABLE
812 THRESHOLD OF TRANSFER RATE OF PORT
813 THRESHOLD OF IOPS OF PORT
814 THRESHOLD OF TRANSFER RATE OF LDEV
815 THRESHOLD OF IOPS OF LDEV
803 ALERT NOTIFICATION TABLE
816 DATE AND TIME
817 ALERT TYPE
818 SYSTEM NAME
819 SUBJECT
820 METRIC TYPE
821 ABNORMAL VALUE
804 SYSTEM MONITORING RESULT TABLE
822 DATE AND TIME
823 SYSTEM NAME
824 THRESHOLD DETERMINATION RESULT OF CPU USAGE OF RDB SERVER
825 THRESHOLD DETERMINATION RESULT OF NETWORK USAGE OF RDB SERVER
826 THRESHOLD DETERMINATION RESULT OF DISK IOPS OF RDB SERVER
827 THRESHOLD DETERMINATION RESULT OF DISK TRANSFER RATE OF RDB SERVER
828 THRESHOLD DETERMINATION RESULT OF IOPS OF STORAGE PORT
829 THRESHOLD DETERMINATION RESULT OF TRANSFER RATE OF STORAGE PORT
805 NODE MONITORING RESULT TABLE
830 DATE AND TIME
831 SYSTEM NAME
832 RDB SERVER NAME
833 NODE NAME
834 PATH OF DEVICE
835 THRESHOLD DETERMINATION RESULT OF IOPS OF DEVICE
836 THRESHOLD DETERMINATION RESULT OF TRANSFER RATE OF DEVICE
837 THRESHOLD DETERMINATION RESULT OF TRANSFER RATE OF LDEV
838 THRESHOLD DETERMINATION RESULT OF IOPS OF LDEV

FIG. 8

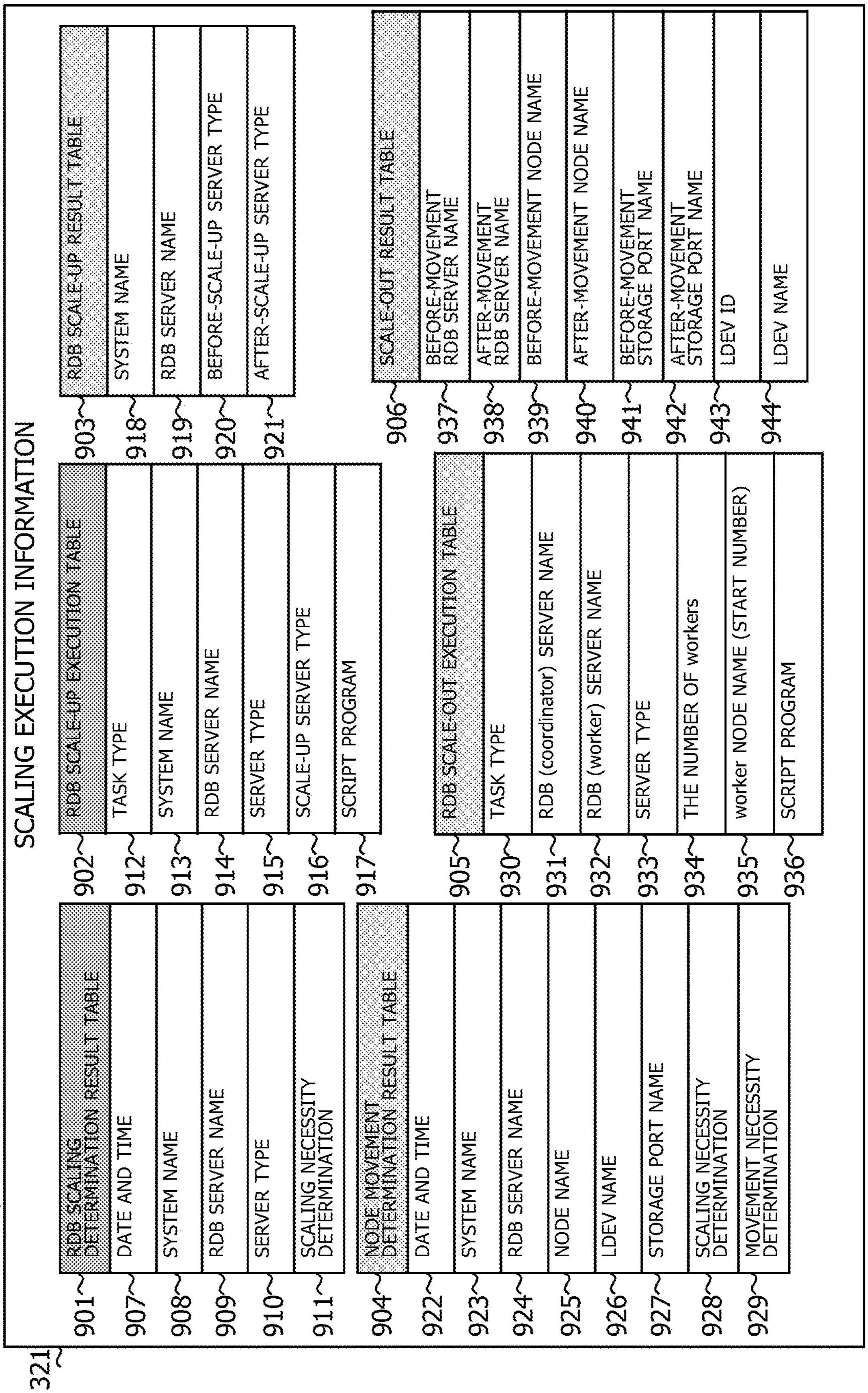
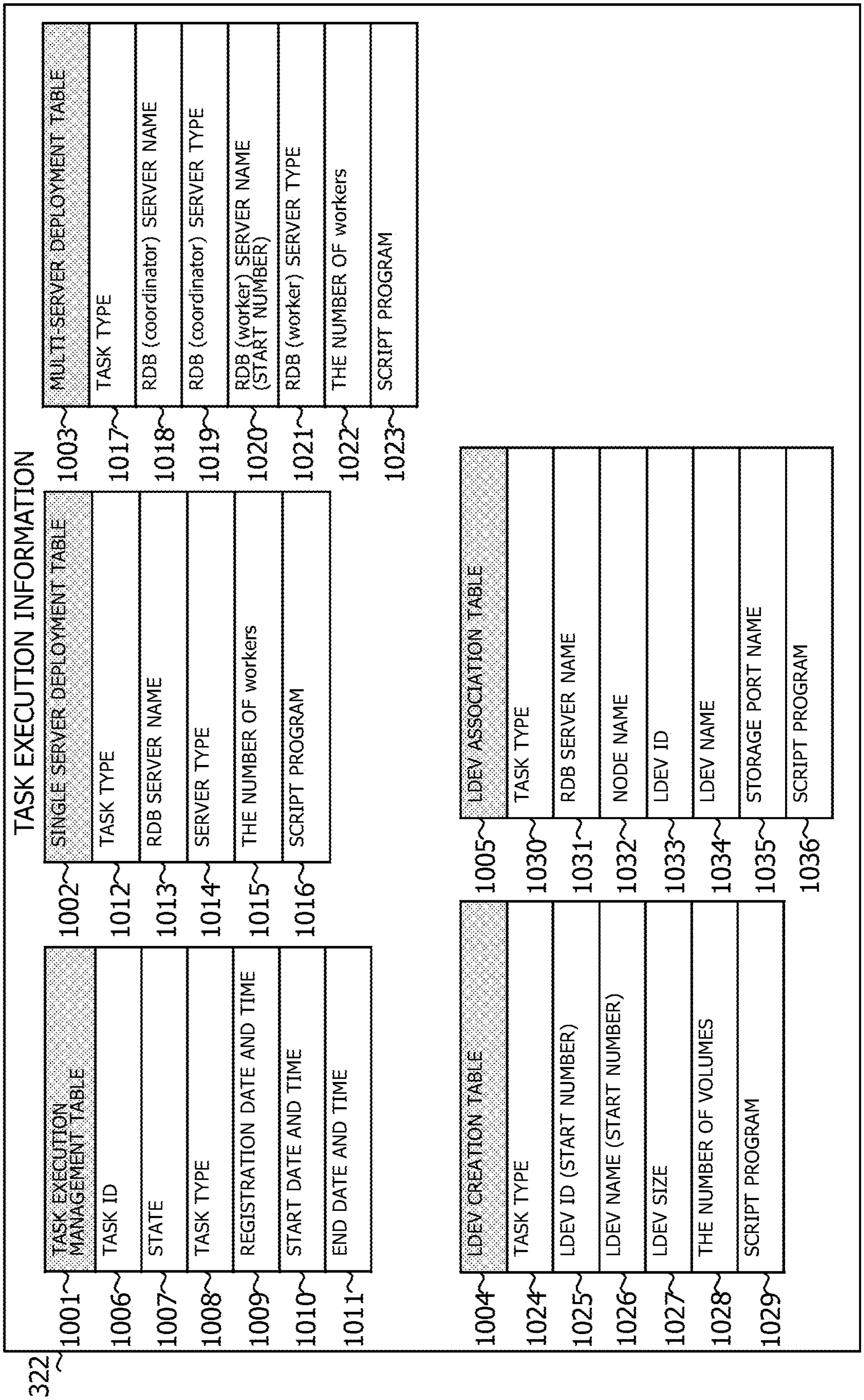

321
SCALING EXECUTION INFORMATION
901 RDB SCALING DETERMINATION RESULT TABLE
907 DATE AND TIME
908 SYSTEM NAME
909 RDB SERVER NAME
910 SERVER TYPE
911 SCALING NECESSITY DETERMINATION
904 NODE MOVEMENT DETERMINATION RESULT TABLE
922 DATE AND TIME
923 SYSTEM NAME
924 RDB SERVER NAME
925 NODE NAME
926 LDEV NAME
927 STORAGE PORT NAME
928 SCALING NECESSITY DETERMINATION
929 MOVEMENT NECESSITY DETERMINATION
902 RDB SCALE-UP EXECUTION TABLE
912 TASK TYPE
913 SYSTEM NAME
914 RDB SERVER NAME
915 SERVER TYPE
916 SCALE-UP SERVER TYPE
917 SCRIPT PROGRAM
905 RDB SCALE-OUT EXECUTION TABLE
930 TASK TYPE
931 RDB (coordinator) SERVER NAME
932 RDB (worker) SERVER NAME
933 SERVER TYPE
934 THE NUMBER OF workers
935 worker NODE NAME (START NUMBER)
936 SCRIPT PROGRAM
903 RDB SCALE-UP RESULT TABLE
918 SYSTEM NAME
919 RDB SERVER NAME
920 BEFORE-SCALE-UP SERVER TYPE
921 AFTER-SCALE-UP SERVER TYPE
906 SCALE-OUT RESULT TABLE
937 BEFORE-MOVEMENT RDB SERVER NAME
938 AFTER-MOVEMENT RDB SERVER NAME
939 BEFORE-MOVEMENT NODE NAME
940 AFTER-MOVEMENT NODE NAME
941 BEFORE-MOVEMENT STORAGE PORT NAME
942 AFTER-MOVEMENT STORAGE PORT NAME
943 LDEV ID
944 LDEV NAME

FIG. 9



322
TASK EXECUTION INFORMATION
1001 TASK EXECUTION MANAGEMENT TABLE
1006 TASK ID
1007 STATE
1008 TASK TYPE
1009 REGISTRATION DATE AND TIME
1010 START DATE AND TIME
1011 END DATE AND TIME
1002 SINGLE SERVER DEPLOYMENT TABLE
1012 TASK TYPE
1013 RDB SERVER NAME
1014 SERVER TYPE
1015 THE NUMBER OF workers
1016 SCRIPT PROGRAM
1003 MULTI-SERVER DEPLOYMENT TABLE
1017 TASK TYPE
1018 RDB (coordinator) SERVER NAME
1019 RDB (coordinator) SERVER TYPE
1020 RDB (worker) SERVER NAME (START NUMBER)
1021 RDB (worker) SERVER TYPE
1022 THE NUMBER OF workers
1023 SCRIPT PROGRAM
1004 LDEV CREATION TABLE
1024 TASK TYPE
1025 LDEV ID (START NUMBER)
1026 LDEV NAME (START NUMBER)
1027 LDEV SIZE
1028 THE NUMBER OF VOLUMES
1029 SCRIPT PROGRAM
1005 LDEV ASSOCIATION TABLE
1030 TASK TYPE
1031 RDB SERVER NAME
1032 NODE NAME
1033 LDEV ID
1034 LDEV NAME
1035 STORAGE PORT NAME
1036 SCRIPT PROGRAM

FIG. 10

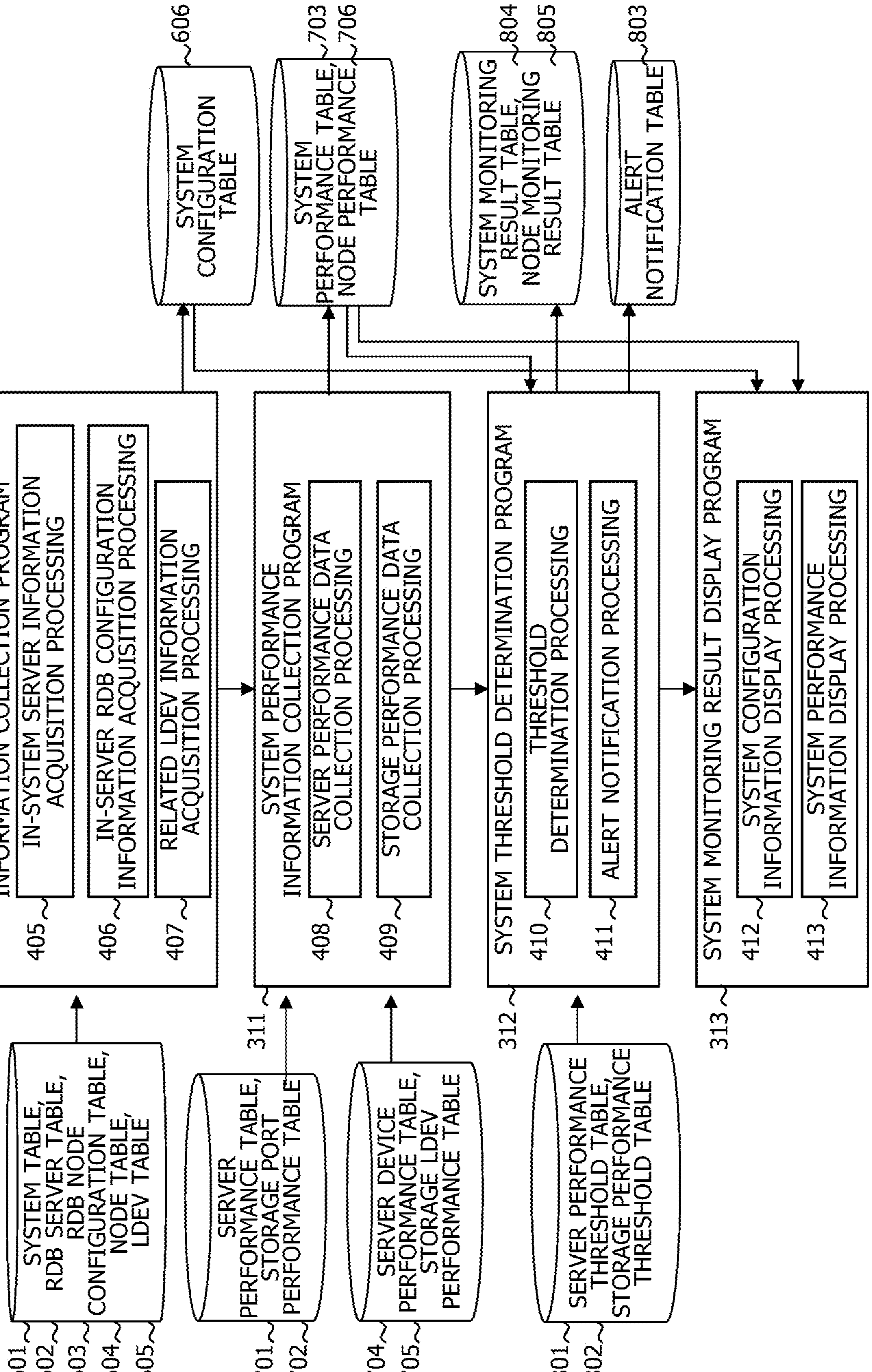
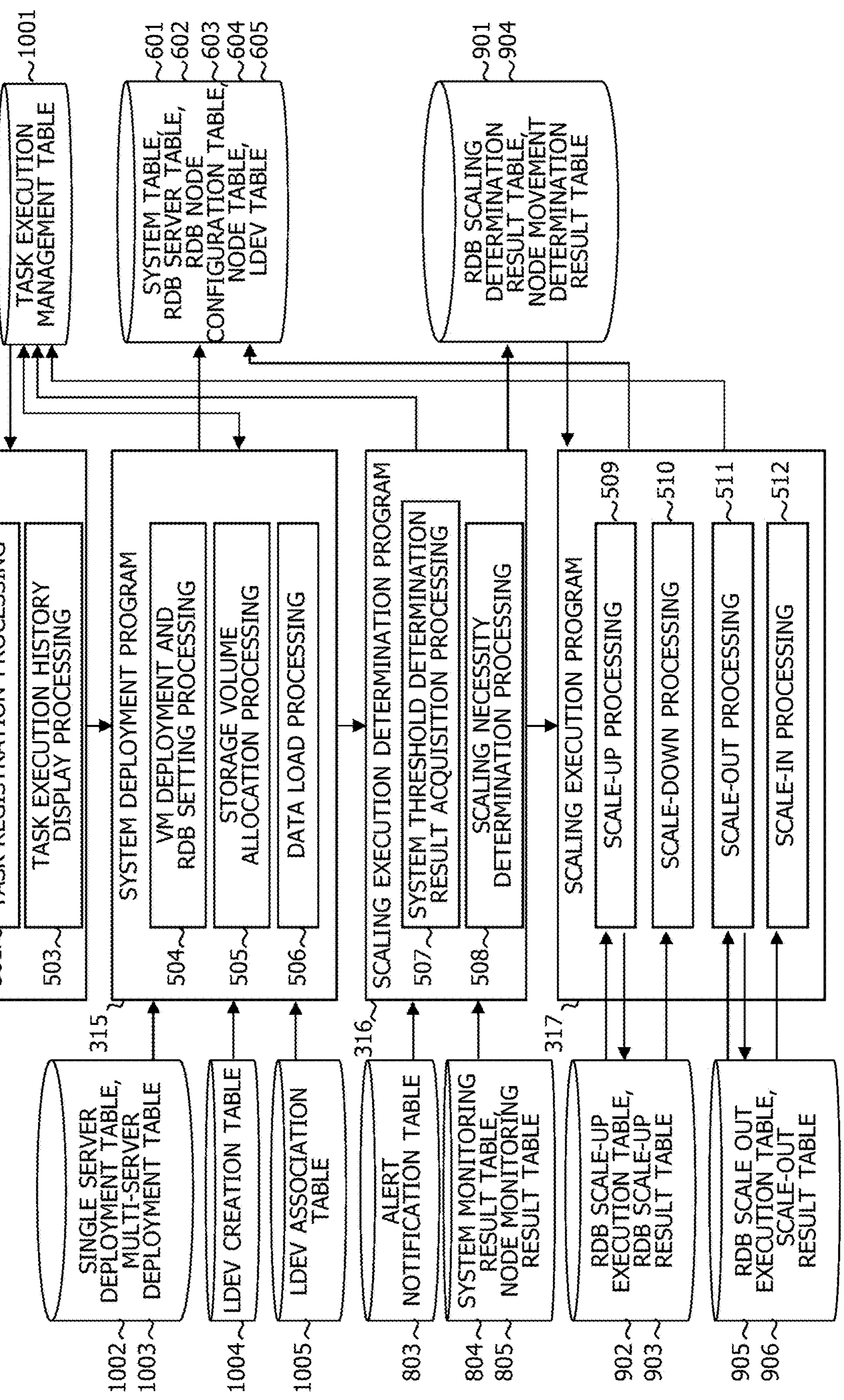

601 SYSTEM TABLE,
602 RDB SERVER TABLE,
603 RDB NODE CONFIGURATION TABLE,
604 NODE TABLE,
605 LDEV TABLE
701 SERVER PERFORMANCE TABLE,
702 STORAGE PORT PERFORMANCE TABLE
704 SERVER DEVICE PERFORMANCE TABLE,
705 STORAGE LDEV PERFORMANCE TABLE
801 SERVER PERFORMANCE THRESHOLD TABLE,
802 STORAGE PERFORMANCE THRESHOLD TABLE
310 SYSTEM CONFIGURATION INFORMATION COLLECTION PROGRAM
405 IN-SYSTEM SERVER INFORMATION ACQUISITION PROCESSING
406 IN-SERVER RDB CONFIGURATION INFORMATION ACQUISITION PROCESSING
407 RELATED LDEV INFORMATION ACQUISITION PROCESSING
311 SYSTEM PERFORMANCE INFORMATION COLLECTION PROGRAM
408 SERVER PERFORMANCE DATA COLLECTION PROCESSING
409 STORAGE PERFORMANCE DATA COLLECTION PROCESSING
312 SYSTEM THRESHOLD DETERMINATION PROGRAM
410 THRESHOLD DETERMINATION PROCESSING
411 ALERT NOTIFICATION PROCESSING
313 SYSTEM MONITORING RESULT DISPLAY PROGRAM
412 SYSTEM CONFIGURATION INFORMATION DISPLAY PROCESSING
413 SYSTEM PERFORMANCE INFORMATION DISPLAY PROCESSING
606 SYSTEM CONFIGURATION TABLE
703 SYSTEM PERFORMANCE TABLE,
706 NODE PERFORMANCE TABLE
804 SYSTEM MONITORING RESULT TABLE,
805 NODE MONITORING RESULT TABLE
803 ALERT NOTIFICATION TABLE

FIG. 11


TASK EXECUTION MANAGEMENT TABLE
1001
SYSTEM TABLE, RDB SERVER TABLE, RDB NODE CONFIGURATION TABLE, NODE TABLE, LDEV TABLE
601
602
603
604
605
RDB SCALING DETERMINATION RESULT TABLE, NODE MOVEMENT DETERMINATION RESULT TABLE
901
904
TASK MANAGEMENT PROGRAM
314
TASK REGISTRATION PROCESSING
501
TASK EXECUTION HISTORY DISPLAY PROCESSING
503
SYSTEM DEPLOYMENT PROGRAM
315
VM DEPLOYMENT AND RDB SETTING PROCESSING
504
STORAGE VOLUME ALLOCATION PROCESSING
505
DATA LOAD PROCESSING
506
SCALING EXECUTION DETERMINATION PROGRAM
316
SYSTEM THRESHOLD DETERMINATION RESULT ACQUISITION PROCESSING
507
SCALING NECESSITY DETERMINATION PROCESSING
508
SCALING EXECUTION PROGRAM
317
SCALE-UP PROCESSING
509
SCALE-DOWN PROCESSING
510
SCALE-OUT PROCESSING
511
SCALE-IN PROCESSING
512
SINGLE SERVER DEPLOYMENT TABLE, MULTI-SERVER DEPLOYMENT TABLE
1002
1003
LDEV CREATION TABLE
1004
LDEV ASSOCIATION TABLE
1005
ALERT NOTIFICATION TABLE
803
SYSTEM MONITORING RESULT TABLE, NODE MONITORING RESULT TABLE
804
805
RDB SCALE-UP EXECUTION TABLE, RDB SCALE-UP RESULT TABLE
902
903
RDB SCALE OUT EXECUTION TABLE, SCALE-OUT RESULT TABLE
905
906

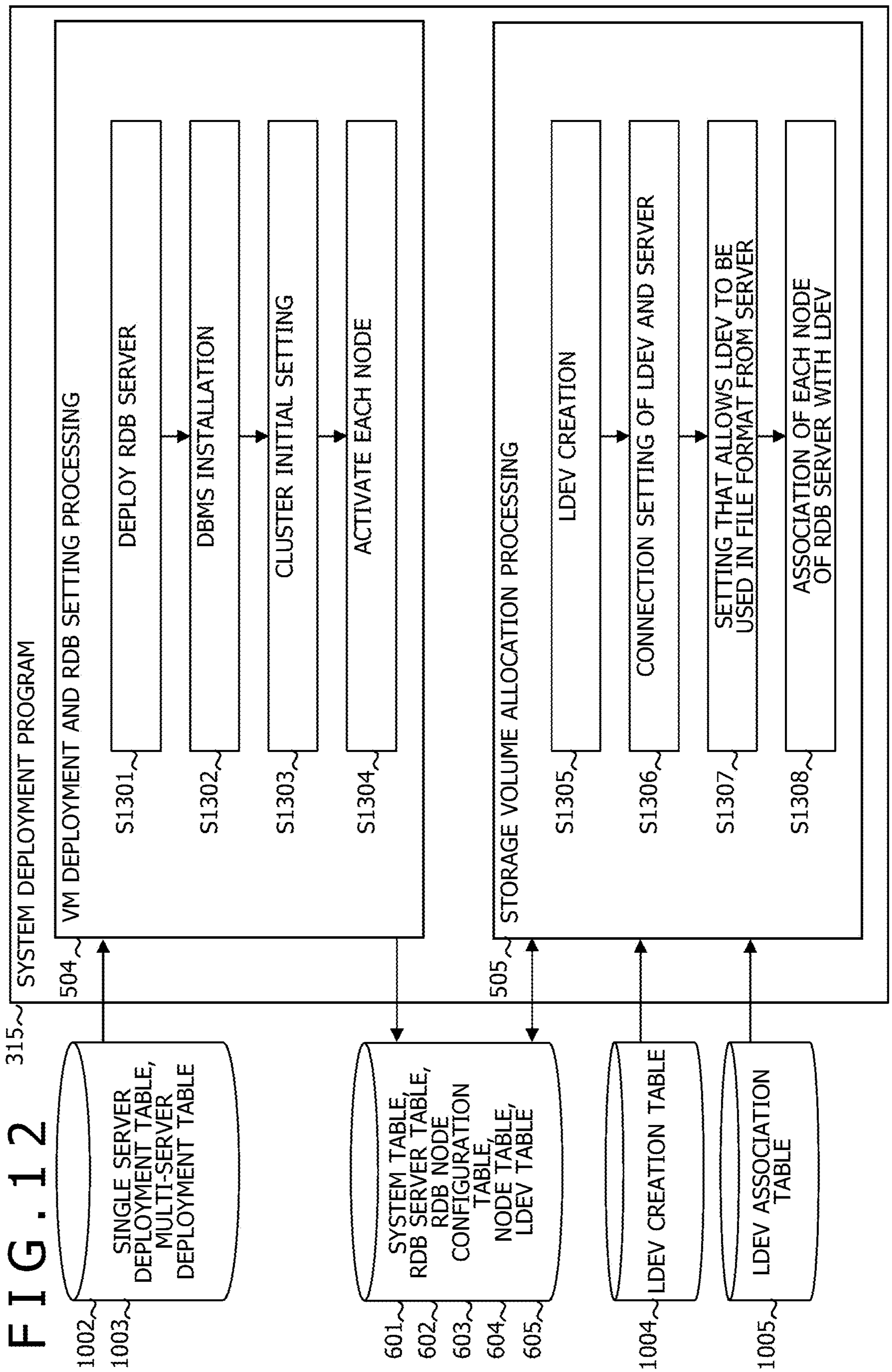

315
SYSTEM DEPLOYMENT PROGRAM
504
VM DEPLOYMENT AND RDB SETTING PROCESSING
S1301
DEPLOY RDB SERVER
S1302
DBMS INSTALLATION
S1303
CLUSTER INITIAL SETTING
S1304
ACTIVATE EACH NODE
505
STORAGE VOLUME ALLOCATION PROCESSING
S1305
LDEV CREATION
S1306
CONNECTION SETTING OF LDEV AND SERVER
S1307
SETTING THAT ALLOWS LDEV TO BE USED IN FILE FORMAT FROM SERVER
S1308
ASSOCIATION OF EACH NODE OF RDB SERVER WITH LDEV
1002
1003
SINGLE SERVER DEPLOYMENT TABLE, MULTI-SERVER DEPLOYMENT TABLE
601
602
603
604
605
SYSTEM TABLE, RDB SERVER TABLE, RDB NODE CONFIGURATION TABLE, NODE TABLE, LDEV TABLE
1004
LDEV CREATION TABLE
1005
LDEV ASSOCIATION TABLE

FIG. 12

FIG. 13

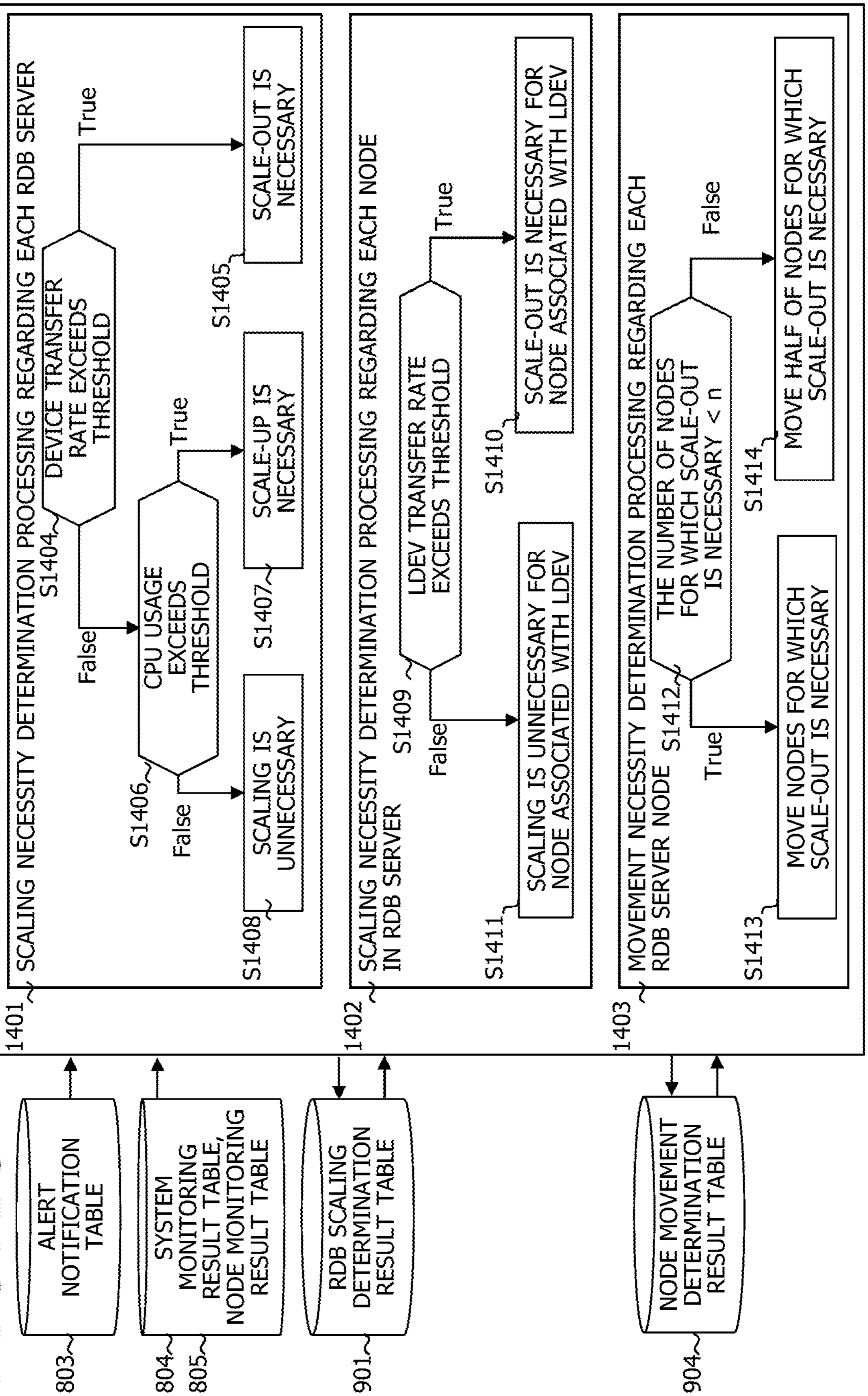

803
ALERT NOTIFICATION TABLE
804
805
SYSTEM MONITORING RESULT TABLE, NODE MONITORING RESULT TABLE
901
RDB SCALING DETERMINATION RESULT TABLE
904
NODE MOVEMENT DETERMINATION RESULT TABLE
508
SCALING NECESSITY DETERMINATION PROCESSING
1401
SCALING NECESSITY DETERMINATION PROCESSING REGARDING EACH RDB SERVER
S1404
DEVICE TRANSFER RATE EXCEEDS THRESHOLD
True
False
S1405
SCALE-OUT IS NECESSARY
S1406
CPU USAGE EXCEEDS THRESHOLD
True
False
S1407
SCALE-UP IS NECESSARY
S1408
SCALING IS UNNECESSARY
1402
SCALING NECESSITY DETERMINATION PROCESSING REGARDING EACH NODE IN RDB SERVER
S1409
LDEV TRANSFER RATE EXCEEDS THRESHOLD
True
False
S1410
SCALE-OUT IS NECESSARY FOR NODE ASSOCIATED WITH LDEV
S1411
SCALING IS UNNECESSARY FOR NODE ASSOCIATED WITH LDEV
1403
MOVEMENT NECESSITY DETERMINATION PROCESSING REGARDING EACH RDB SERVER NODE
S1412
THE NUMBER OF NODES FOR WHICH SCALE-OUT IS NECESSARY < n
False
True
S1414
MOVE HALF OF NODES FOR WHICH SCALE-OUT IS NECESSARY
S1413
MOVE NODES FOR WHICH SCALE-OUT IS NECESSARY

FIG. 14

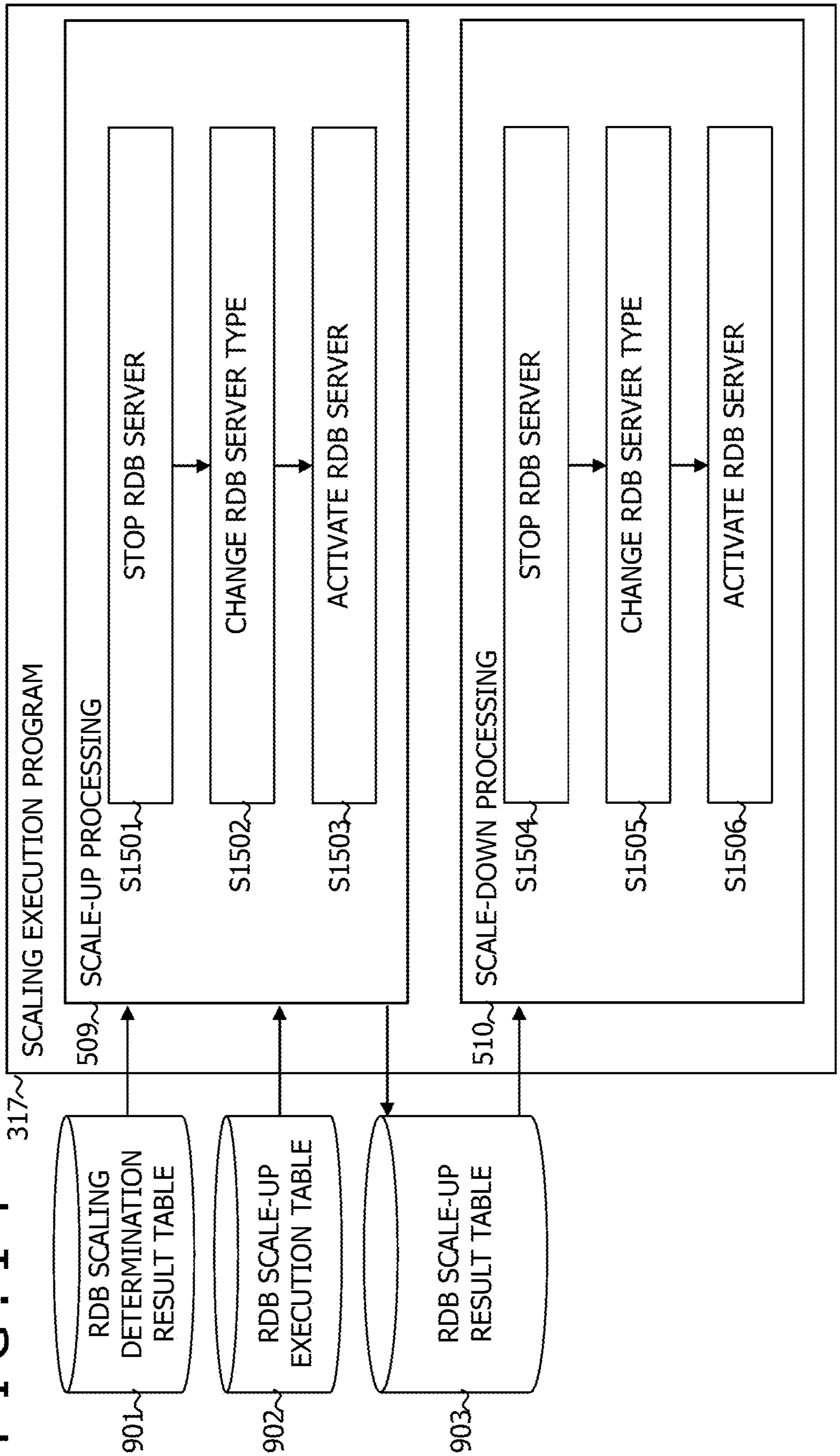

317
SCALING EXECUTION PROGRAM
509
SCALE-UP PROCESSING
S1501
STOP RDB SERVER
S1502
CHANGE RDB SERVER TYPE
S1503
ACTIVATE RDB SERVER
510
SCALE-DOWN PROCESSING
S1504
STOP RDB SERVER
S1505
CHANGE RDB SERVER TYPE
S1506
ACTIVATE RDB SERVER
901
RDB SCALING DETERMINATION RESULT TABLE
902
RDB SCALE-UP EXECUTION TABLE
903
RDB SCALE-UP RESULT TABLE

FIG. 15

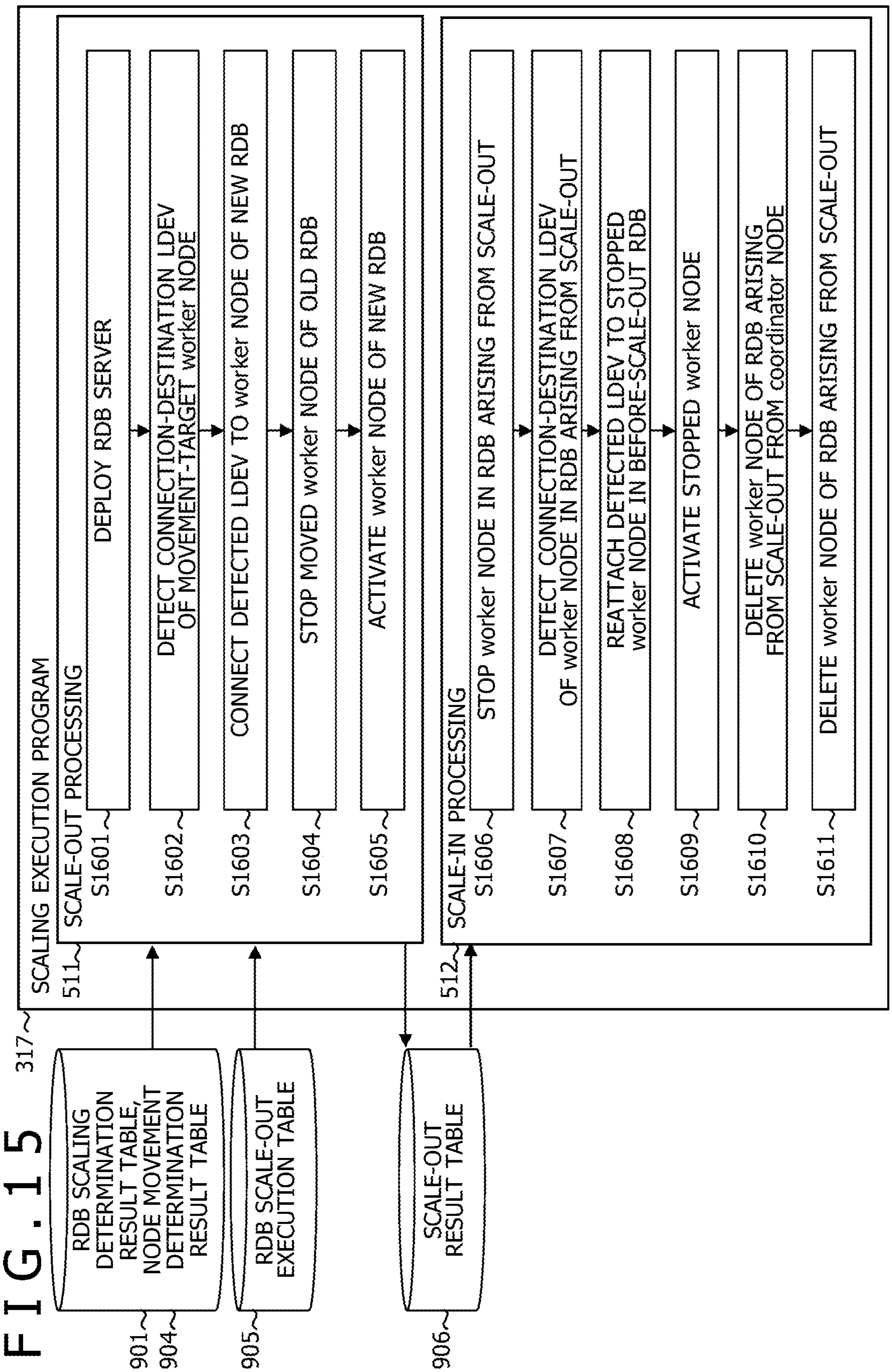

317
SCALING EXECUTION PROGRAM
511
SCALE-OUT PROCESSING
S1601 DEPLOY RDB SERVER
S1602 DETECT CONNECTION-DESTINATION LDEV OF MOVEMENT-TARGET worker NODE
S1603 CONNECT DETECTED LDEV TO worker NODE OF NEW RDB
S1604 STOP MOVED worker NODE OF OLD RDB
S1605 ACTIVATE worker NODE OF NEW RDB
512
SCALE-IN PROCESSING
S1606 STOP worker NODE IN RDB ARISING FROM SCALE-OUT
S1607 DETECT CONNECTION-DESTINATION LDEV OF worker NODE IN RDB ARISING FROM SCALE-OUT
S1608 REATTACH DETECTED LDEV TO STOPPED worker NODE IN BEFORE-SCALE-OUT RDB
S1609 ACTIVATE STOPPED worker NODE
S1610 DELETE worker NODE OF RDB ARISING FROM SCALE-OUT FROM coordinator NODE
S1611 DELETE worker NODE OF RDB ARISING FROM SCALE-OUT
901
904
RDB SCALING DETERMINATION RESULT TABLE, NODE MOVEMENT DETERMINATION RESULT TABLE
905
RDB SCALE-OUT EXECUTION TABLE
906
SCALE-OUT RESULT TABLE

FIG. 16

1701

| TASK ID | STATE | TYPE | START DATE AND TIME | END DATE AND TIME |
|---|---|---|---|---|
| TASK 1 | NORMAL | SYSTEM DEPLOYMENT | t0 | t0 + a1 |
| TASK 2 | NORMAL | SCALE-OUT | t2 | t2 + a2 |
| TASK 3 | NORMAL | SCALE-IN | t4 | t4 + a3 |

1702

| SYSTEM | RDB NAME | Worker NAME | DEVICE PATH | LDEV NAME | STORAGE PORT |
|---|---|---|---|---|---|
| App01 | VM01 | worker1 | /mnt/disk1 | LDEV1 | CL1-A |
| App01 | VM01 | worker2 | /mnt/disk2 | LDEV2 | CL1-A |

1703

| SYSTEM | RDB NAME | Worker NAME | DEVICE PATH | LDEV NAME | STORAGE PORT |
|---|---|---|---|---|---|
| App01 | VM01 | worker1 | /mnt/disk1 | LDEV1 | CL1-A |
| App01 | VM02 | worker3 | /mnt/disk3 | LDEV2 | CL2-A |

FIG. 17

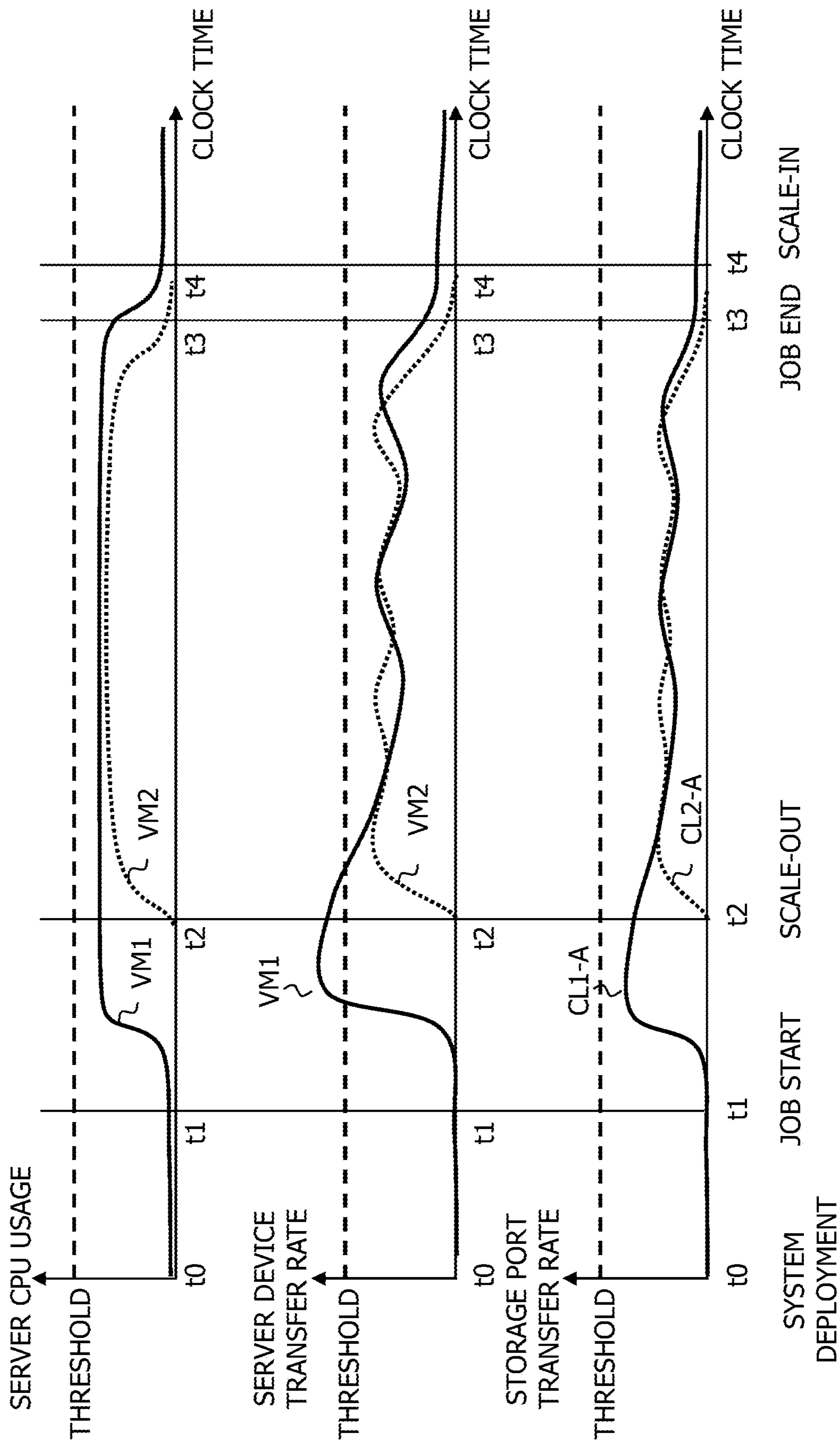
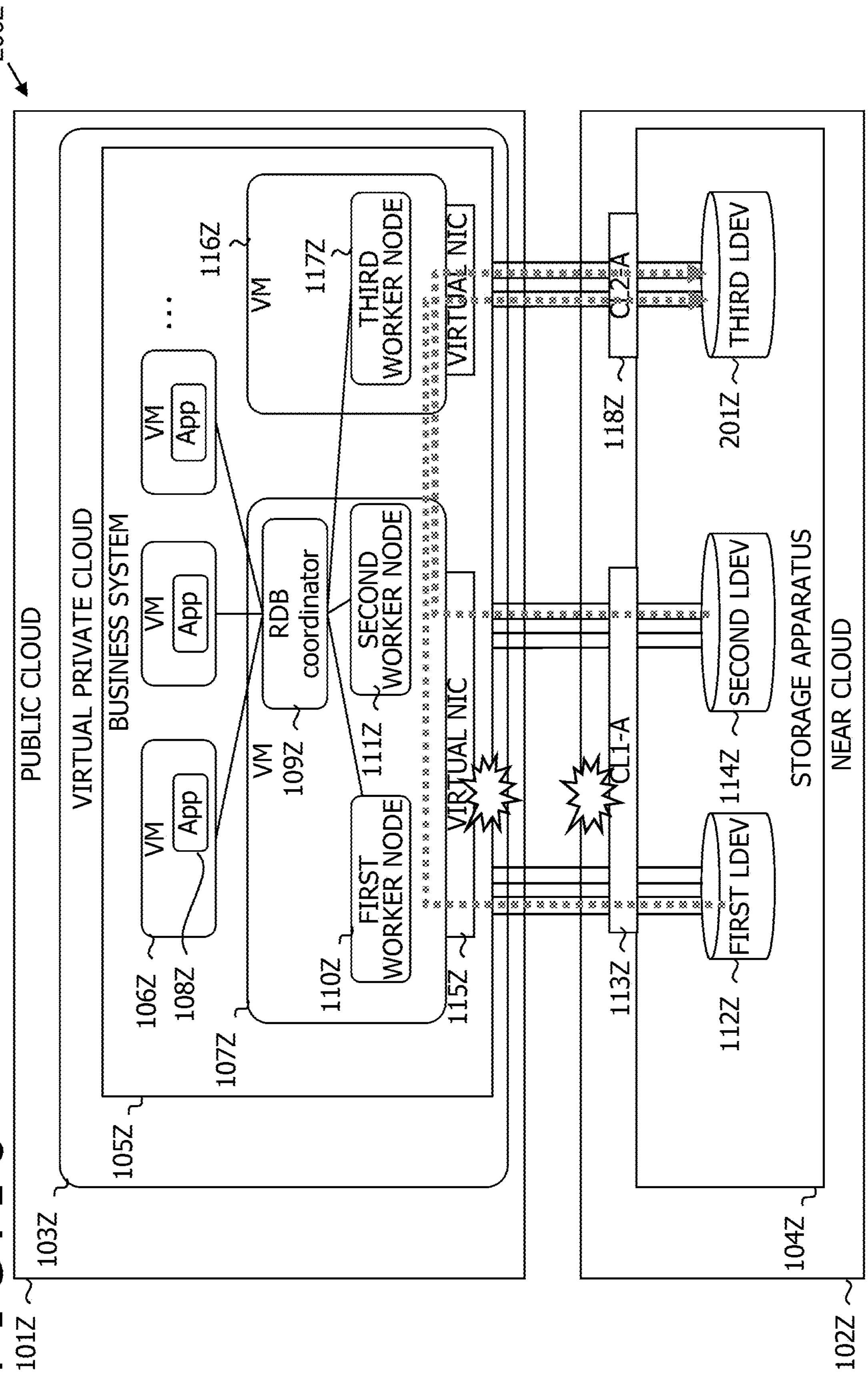

SERVER CPU USAGE
THRESHOLD
VM1
VM2
CLOCK TIME
t0
t1
t2
t3
t4
SERVER DEVICE TRANSFER RATE
THRESHOLD
VM1
VM2
CLOCK TIME
t0
t1
t2
t3
t4
STORAGE PORT TRANSFER RATE
THRESHOLD
CL1-A
CL2-A
CLOCK TIME
t0
t1
t2
t3
t4
SYSTEM DEPLOYMENT
JOB START
SCALE-OUT
JOB END
SCALE-IN

FIG. 18



200Z
101Z
PUBLIC CLOUD
103Z
VIRTUAL PRIVATE CLOUD
105Z
BUSINESS SYSTEM
106Z
VM
App
108Z
VM
App
VM
App
...
107Z
VM
109Z
RDB coordinator
110Z
FIRST WORKER NODE
111Z
SECOND WORKER NODE
116Z
VM
117Z
THIRD WORKER NODE
115Z
VIRTUAL NIC
VIRTUAL NIC
102Z
NEAR CLOUD
104Z
STORAGE APPARATUS
113Z
CL1-A
118Z
CL2-A
112Z
FIRST LDEV
114Z
SECOND LDEV
201Z
THIRD LDEV

# SCALING SYSTEM AND CALCULATION METHOD

## BACKGROUND OF THE INVENTION

### 1. Field of the Invention

The present invention relates to a scaling system and a calculation method.

### 2. Description of the Related Art

A cloud bursting technique is known in which calculation resources of a public cloud are temporarily used in response to a sudden increase in the traffic in monthly processing or the like. The workload for which the cloud bursting is required is unsteady and therefore preliminary design is difficult. JP-2015-504218-T (hereinafter, patent document 1), a system including a master node and plural worker nodes controlled by the master node is disclosed. In this system, each worker node stores 25 or more module-system blocks of a distributed database and each module-system block has a size of 5 Gbytes or larger and has an associated log file.

## SUMMARY OF THE INVENTION

In the invention described in patent document 1, it takes a long time to start cloud bursting processing.

A scaling system according to a first aspect of the present invention is a scaling system including a first base in which first calculating apparatus in which a first worker node and a second worker node operate is set and a second base in which storage apparatus connected to the first base by a network is set. The scaling system is configured to range over the first base and the second base. The storage apparatus includes a first network port and a second network port. The storage apparatus includes a first volume accessed by the first worker node and a second volume accessed by the second worker node. Second calculating apparatus is further set in the first base. The second worker node is moved to the second calculating apparatus and is caused to operate as a third worker node and the second volume is caused to communicate with the third worker node through the second network port if the transfer rate of the first calculating apparatus or the transfer rate of the first network port exceeds a predetermined threshold when the first volume and the second volume are executing communication with the first calculating apparatus through the first network port.

A calculation method according to a second aspect of the present invention is a calculation method executed by a computer included in a scaling system configured to range over a first base in which first calculating apparatus in which a first worker node and a second worker node operate is set and a second base in which storage apparatus connected to the first base by a network is set. The storage apparatus includes a first network port and a second network port. The storage apparatus includes a first volume accessed by the first worker node and a second volume accessed by the second worker node. Second calculating apparatus is further set in the first base. The calculation method includes moving the second worker node to the second calculating apparatus to cause the second worker node to operate as a third worker node and causing the second volume to communicate with the third worker node through the second network port if the transfer rate of the first calculating apparatus exceeds a predetermined threshold when the first volume and the second volume are executing communication with the first calculating apparatus through the first network port.

According to the present invention, cloud bursting processing can be rapidly started.

## BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is an outline diagram of a scaling system;

FIG. **2** is an overall configuration diagram of the scaling system;

FIG. **3** is a configuration diagram of a monitoring server;

FIG. **4** is a configuration diagram of an operations automation server;

FIG. **5** is a diagram illustrating details of system configuration information;

FIG. **6** is a diagram illustrating details of system performance information;

FIG. **7** is a diagram illustrating details of system monitoring information;

FIG. **8** is a diagram illustrating details of scaling execution information;

FIG. **9** is a diagram illustrating details of task execution information;

FIG. **10** is a diagram illustrating monitoring processing executed by the monitoring server;

FIG. **11** is a diagram illustrating automation processing executed by the operations automation server;

FIG. **12** is a diagram illustrating processing executed by a system deployment program;

FIG. **13** is a diagram illustrating details of scaling necessity determination processing executed by a scaling execution determination program;

FIG. **14** is a diagram illustrating scale-up processing and scale-down processing executed by a scaling execution program;

FIG. **15** is a diagram illustrating scale-out processing and scale-in processing executed by the scaling execution program;

FIG. **16** is a screen display example of an execution result illustrating change in the system performance;

FIG. **17** is a screen display example of the execution result illustrating the change in the system performance; and

FIG. **18** is a diagram illustrating one example of scale-out processing in a configuration of a related art.

## DESCRIPTION OF THE PREFERRED EMBODIMENT

### First Embodiment

A first embodiment of the scaling system according to the present invention will be described below with reference to FIG. **1** to FIG. **18**.

FIG. **1** is an outline diagram of a scaling system **100**. The scaling system **100** is a hybrid cloud configuration that ranges over a public cloud **101** and a near cloud **102** and properly scales out a calculation system S including a database and a storage as described later. A virtual cloud **307** for management and a virtual cloud **103** for business are included in the public cloud **101**. The management virtual cloud **307** and the business virtual cloud **103** are both a virtual private cloud. Storage apparatus **104** is included in the near cloud **102**. The storage apparatus **104** includes one or more logical devices (LDEV). Hereinafter, the public cloud **101** will be referred to also as "first base" and the near cloud **102** will be referred to also as "second base."

A business system **105** is included in the virtual cloud **103** for business. The business system **105** includes an application server (virtual machine, VM) **106** that is a virtual machine and a database server **107**. An application (App) **108** is installed on the application server **106**. A distributed relational database (RDB) is installed on the database server **107**. This database includes one coordinator node **109**, a first worker node **110**, and a second worker node **111**.

The first worker node **110** and a first LDEV **112** in the storage apparatus **104** are coupled by plural connections via a first storage port **113**. Similarly, the second worker node **111** and a second LDEV **114** are coupled by plural connections via the first storage port **113**. Hereinafter, a server in which at least either the coordinator nodes or the worker nodes are operating will be referred to as "database server." In the drawing, the first storage port **113** is described as "CL1-A" and a second storage port **118** is described as "CL2-A." Hereinafter, the first worker node **110** will be referred to also as "first calculating apparatus" and the second worker node **111** will be referred to also as "second calculating apparatus."

Here, when either the I/O transfer rate of a virtual NIC **115** of the database server **107** or the transfer rate of the first storage port **113** exceeds a threshold, the following processing is executed in the present embodiment. Specifically, a database server **116** is newly created and a third worker node **117** is created in this database server **116** and is added to the coordinator node **109**. The third worker node **117** and the second LDEV **114** are coupled by plural connections via the second storage port **118**.

FIG. **2** is an overall configuration diagram of the scaling system **100**. The scaling system **100** includes the public cloud **101** and the near cloud **102** as described above. The public cloud **101** includes the virtual cloud **307** for management and the virtual cloud **103** for business. In the virtual cloud **103** for business, the business system **105** and a calculation-side collection tool **302** that collects configuration information of the system and performance information of the server are included. The business system **105** is connected to the storage apparatus **104** in the near cloud **102** by a dedicated line **301**.

The near cloud **102** includes a storage management server **303** and a tenant storage rack **104**A including the storage apparatus **104**. The storage management server **303** has a setting tool **304** to make settings for the storage apparatus **104** and a storage-side collection tool **305** that collects configuration information and storage performance information of the storage apparatus **104**. An operations manager who manages the scaling system **100** accesses the virtual cloud **307** for management in the public cloud **101** by using a client PC **306**.

In the virtual cloud **307** for management, a monitoring server **308**, an operations automation server **309**, system configuration information **318**, system performance information **319**, system monitoring information **320**, scaling execution information **321**, and task execution information **322** are included. The monitoring server **308** has a system configuration information collection program **310**, a system performance information collection program **311**, a system threshold determination program **312**, and a system monitoring result display program **313**. In the operations automation server **309**, a task management program **314**, a system deployment program **315**, a scaling execution determination program **316**, and a scaling execution program **317** are included.

Data used by the monitoring server **308** and the operations automation server **309** are stored in the system configuration information **318**, the system performance information **319**, the system monitoring information **320**, the scaling execution information **321**, and the task execution information **322**.

FIG. **3** is a configuration diagram of the monitoring server **308**. The monitoring server **308** includes a monitoring server memory **401**A that is readable-writable storing apparatus, a monitoring server processor **402**A that is a central processing unit, monitoring server storing apparatus **403**A that is non-volatile storing apparatus, and a monitoring server communication interface **404**A. The monitoring server **308** deploys plural programs stored in the monitoring server storing apparatus **403**A or a ROM that is not illustrated in the diagram in an automation server memory **401**B and executes the programs. That is, the system configuration information collection program **310**, the system performance information collection program **311**, the system threshold determination program **312**, and the system monitoring result display program **313** are implemented by using the automation server memory **401**B, an automation server processor **402**B, automation server storing apparatus **403**B, and an automation server communication interface **404**B.

In the monitoring server storing apparatus **403**A, the system configuration information **318**, the system performance information **319**, and the system monitoring information **320** are stored. Input information from the calculation-side collection tool **302** in the virtual cloud **103** for business and the storage-side collection tool **305** in the storage management server **303** and input-output information exchanged from and to the client PC **306** via a Web browser are input and output to and from the monitoring server communication interface **404**A.

In the system configuration information collection program **310**, in-system server information acquisition processing **405**, in-server RDB configuration information acquisition processing **406**, and related LDEV information acquisition processing **407** are included. The in-system server information acquisition processing **405** is processing of acquiring configuration information of all servers included in the scaling system **100**, for example, information such as the names of the servers and instance IDs of the servers. The in-server RDB configuration information acquisition processing **406** is processing of acquiring configuration information of the relational database (RDB) included in the virtual cloud **103** for business, for example, information such as IP addresses of worker nodes. The related LDEV information acquisition processing **407** is processing of acquiring information on all logical devices included in the scaling system **100**, for example, information such as paths and LDEV IDs of the devices.

In the system performance information collection program **311**, server performance data collection processing **408** and storage performance data collection processing **409** are included. The server performance data collection processing **408** is processing of acquiring dynamic performance information of all database servers included in the scaling system **100**, for example, information such as the CPU usage and the disk transfer rate. The storage performance data collection processing **409** is processing of acquiring dynamic performance information of all pieces of storage apparatus **104** included in the scaling system **100**, for example, information such as the transfer rate of the storage port.

In the system threshold determination program **312**, threshold determination processing **410** and alert notification processing **411** are included. The threshold determination processing **410** is processing of executing abnormality sensing by using an evaluation subject value and a threshold and storing the processing result. In the threshold determination processing **410**, for example, comparison between a threshold **806** of the CPU usage and a CPU usage **710** to be described later is carried out and the comparison result is stored in a threshold determination result **824** of the CPU usage of the RDB server. The alert notification processing **411** is processing of issuing an alert notification on the basis of the processing result of the threshold determination processing **410**.

In the system monitoring result display program **313**, system configuration information display processing **412** and system performance information display processing **413** are included. The system configuration information display processing **412** is processing of displaying configuration information of the scaling system **100** on output apparatus, for example, a liquid crystal display. The system performance information display processing **413** is processing of displaying performance information of the scaling system **100** on output apparatus, for example, a liquid crystal display. The output apparatus deemed as the processing subject by the system configuration information display processing **412** and the system performance information display processing **413** may be output apparatus included in the monitoring server **308** or may be output apparatus included in the client PC **306**.

FIG. **4** is a configuration diagram of the operations automation server **309**. The operations automation server **309** includes the automation server memory **401**B that is readable-writable storing apparatus, the automation server processor **402**B that is a central processing unit, the automation server storing apparatus **403**B that is non-volatile storing apparatus, and the automation server communication interface **404**B. The monitoring server **308** deploys plural programs stored in the monitoring server storing apparatus **403**A or a ROM that is not illustrated in the diagram in the automation server memory **401**B and executes the programs. That is, the task management program **314**, the system deployment program **315**, the scaling execution determination program **316**, and the scaling execution program **317** are implemented by using the automation server memory **401**B, the automation server processor **402**B, the automation server storing apparatus **403**B, and the automation server communication interface **404**B.

In the automation server storing apparatus **403**B, the system monitoring information **320**, the scaling execution information **321**, and the task execution information **322** are stored. Input-output information from and to the business system **105** in the virtual cloud **103** for business and the setting tool **304** in the storage management server **303** and input-output information exchanged from and to the client PC **306** via a Web browser are input and output to and from the automation server communication interface **404**B.

In the task management program **314**, task registration processing **501**, task execution processing **502**, and task execution history display processing **503** are included. The task registration processing **501** is processing of registering, as a task, deployment of the scaling system **100** ordered from the client PC **306**, scale-out of the scaling system **100** ordered from scaling necessity determination processing **508** to be described later, and so forth. The task execution processing **502** is processing of executing the registered task and calls the scaling execution program **317** according to need. For example, when scale-out is registered as a task, the task execution processing **502** calls and executes scale-out processing **511** to be described later. The task execution history display processing **503** is processing of displaying the execution history of the task on output apparatus.

In the system deployment program **315**, VM deployment and RDB setting processing **504**, storage volume allocation processing **505**, and data load processing **506** are included. The VM deployment and RDB setting processing **504** is processing of executing deployment of a virtual machine and setting of the RDB. The storage volume allocation processing **505** is processing of deciding the correspondence relation between LDEVs of the storage apparatus **104** and worker nodes. The data load processing **506** is processing of causing each LDEV of the storage apparatus **104** to read in necessary information.

The scaling execution determination program **316** executes system threshold determination result acquisition processing **507** and the scaling necessity determination processing **508**. The system threshold determination result acquisition processing **507** is processing of acquiring the processing result of the threshold determination processing **410** of the system threshold determination program **312**. The scaling necessity determination processing **508** is processing of determining whether or not scale-up, scale-down, scale-out, and scale-in are necessary.

The scaling execution program **317** executes scale-up processing **509**, scale-down processing **510**, the scale-out processing **511**, and scale-in processing **512**. The scale-up processing **509** is processing of scaling up the business system **105**. The scale-down processing **510** is processing of scaling down the business system **105**. The scale-out processing **511** is processing of scaling out the business system **105**. The scale-in processing **512** is processing of scaling in the business system **105**.

Details of the system configuration information **318**, the system performance information **319**, the system monitoring information **320**, the scaling execution information **321**, and the task execution information **322** will be described with reference to FIG. **5** to FIG. **9**. Each of the system configuration information **318**, the system performance information **319**, the system monitoring information **320**, the scaling execution information **321**, and the task execution information **322** includes plural tables as described below. However, the following description is one example and the storage system of the information is not limited to the table format. Furthermore, in the following, when the same kind of information is included in plural tables, description is omitted regarding the second and subsequent appearances of the information.

FIG. **5** is a diagram illustrating details of the system configuration information **318**. The system configuration information **318** is composed of a system table **601**, an RDB server table **602**, an RDB node configuration table **603**, a node table **604**, an LDEV table **605**, and a system configuration table **606**. The system configuration information **318** is automatically generated by processing to be described later. The system table **601**, the RDB server table **602**, and the system configuration table **606** are created as each one table, and records corresponding to the number of RDB servers are added to each table every date and time of information collection. The RDB node configuration table **603** and the node table **604** are created as each one table, and records corresponding to the number of nodes are added to each table every date and time of information collection. To the LDEV table **605**, the same number of records as the number of volumes (LDEV) of the storage apparatus **104** are added every date and time of information collection.

Each record of the system table **601** is composed of fields of date and time **607**, a system name **608**, and an RDB server name **609**. The date and time **607** is information on date and clock time that serves as a criterion for determining whether or not information stored in the system table **601** is new or old. The system name **608** is the name of the calculation system S controlled by the scaling system **100**. Since the calculation system S corresponds to the application **108** in the business system **105**, the system name **608** may be read as the name of the application **108**. The RDB server name **609** is an identifier to identify the database server (VM) included in the calculation system S. For example, in the example illustrated in FIG. **1**, the database server **107** or the VM **116** is set in the RDB server name **609**.

Each record of the RDB server table **602** is composed of fields of date and time **610**, a system name **611**, an RDB server name **612**, an RDB server type **613**, an RDB instance ID **614**, an IP address **615** of the RDB, and a storage port name **616**.

The date and time **610** is information on date and clock time that serves as a criterion for determining whether or not information stored in the RDB server table **602** is new or old. The system name **611** and the RDB server name **612** are the same kind of information as the system name **608** and the RDB server name **609**, respectively. Hereinafter, when the name of information is the same, description showing that the information is the same kind of information is also not made. The RDB server type **613** is identification information of resources allocated to the database server. For example, this identification information is "minimum configuration" that is a combination of a CPU with four cores and a memory of 8 GB, "memory-increased configuration" that is a combination of a CPU with four cores and a memory of 16 GB, "maximum configuration" that is a combination of a CPU with 32 cores and a memory of 64 GB, or the like.

The RDB instance ID **614** is the instance ID of the database server **107**. The IP address **615** of the RDB is the IP address of the database server **107**. The storage port name **616** is the identifiers of all ports that are ports that the storage apparatus **104** has and are used by the calculation system S.

Each record of the RDB node configuration table **603** is composed of fields of date and time **617**, an RDB server name **618**, a node name **619**, and a node type **620**. The node name **619** is the name of the node, for example, "first worker node." The node type **620** is the type of the node, for example, "worker node" or "coordinator node."

Each record of the node table **604** is composed of fields of date and time **621**, an RDB server name **622**, a node name **623**, a node type **624**, an IP address **625** of the node, and a path **626** of the node, and a port number **627** of the node. The IP address **625** of the node is the IP address of the server in which the node operates. The path **626** of the node is a so-called directory in a virtual machine. For example, a certain storage volume looks to be "/dev/sda1" from the server and "/dev/sda1" is used in such a manner as to be mounted on a directory "/mnt/disk1" of the first worker node **110**. The port number **627** of the node is the storage port used for accessing the storage volume in which data of the node is stored from the server, for example, "CL1-A" or the like.

Each record of the LDEV table **605** is composed of fields of date and time **628**, a node name **629**, a path **630** of the device, an LDEV ID **631**, an LDEV name **632**, a storage port name **633**, and an LUN ID **634**. The date and time **628** is information on date and clock time that serves as a criterion for determining whether or not information stored in the LDEV table **605** is new or old. The path **630** of the device is information showing the logical position of the LDEV and is, for example, "/mnt/disk1." The LDEV ID **631** is the identifier of the LDEV. The LDEV name **632** is the name of the LDEV. The LUN ID **634** is the logical device number (logical unit number) of the LDEV.

Each record of the system configuration table **606** is composed of fields of date and time **635**, a system name **636**, an RDB server name **637**, a node name **638**, an LDEV name **639**, and a storage port name **640**. The date and time **635** is information on date and clock time that serves as a criterion for determining whether or not information stored in the system configuration table **606** is new or old.

FIG. **6** is a diagram illustrating details of the system performance information **319**. The system performance information **319** is composed of a server performance table **701**, a storage port performance table **702**, a system performance table **703**, a server device performance table **704**, a storage LDEV performance table **705**, and a node performance table **706**. For the system performance information **319**, an operating system of each of the database server and the storage apparatus **104** updates the respective items. As each of the server performance table **701**, the system performance table **703**, and the server device performance table **704**, as many tables as the number of database servers are generated. As the storage port performance table **702**, as many tables as the number of storage ports that the storage apparatus **104** has are generated. As the storage LDEV performance table **705**, as many tables as the number of LDEVs generated in the storage apparatus **104** are generated.

Each record of the server performance table **701** is composed of fields of date and time **708**, an RDB server name **709**, a CPU usage **710**, a network usage **711**, a disk transfer rate **712**, and a disk IOPS **713**. The CPU usage **710**, the network usage **711**, the disk transfer rate **712**, and the disk IOPS **713** are the CPU usage, the network usage, the disk transfer rate, and the disk IOPS of the server that change as needed.

Each record of the storage port performance table **702** is composed of fields of date and time **714**, a storage name **715**, a storage port name **716**, a transfer rate **717** of the storage port, an IOPS **718** of the storage port, and a response time **719** of the storage port. The date and time **714** is information on date and clock time that serves as a criterion for determining whether or not information stored in the storage port performance table **702** is new or old. The storage name **715** is the name of the storage apparatus **104** including the storage port. The transfer rate **717** of the storage port, the IOPS **718** of the storage port, and the response time **719** of the storage port are the transfer rate, the IOPS, and the response speed of the storage port that change as needed.

Each record of the system performance table **703** is composed of fields of date and time **720**, a system name **721**, a CPU usage **722** of the RDB server, a network usage **723** of the RDB server, a disk IOPS **724** of the RDB server, a disk transfer rate **725** of the RDB server, an IOPS **726** of the storage port, a transfer rate **727** of the storage port, and a response time **728** of the storage port. The CPU usage **722** of the RDB server, the network usage **723** of the RDB server, the disk IOPS **724** of the RDB server, and the disk transfer rate **725** of the RDB server are the same kind of information as the CPU usage **710**, the network usage **711**, the disk IOPS **713**, and the disk transfer rate **712**, respectively.

Each record of the server device performance table **704** is composed of fields of date and time **729**, an RDB server name **730**, a path **731** of the device, a transfer rate **732** of the device, and an IOPS **733** of the device. Each record of the storage LDEV performance table **705** is composed of fields of date and time **734**, a storage name **735**, an LDEV ID **736**, a transfer rate **737** of the LDEV, an IOPS **738** of the LDEV, and a response time **739** of the LDEV.

Each record of the node performance table **706** is composed of fields of date and time **740**, a system name **741**, an RDB server name **742**, a node name **743**, a path **744** of the device, an IOPS **745** of the device, a transfer rate **746** of the device, a transfer rate **747** of the LDEV, an IOPS **748** of the LDEV, and a response time **749** of the LDEV.

FIG. **7** is a diagram illustrating details of the system monitoring information **320**. The system monitoring information **320** is composed of a server performance threshold table **801**, a storage performance threshold table **802**, an alert notification table **803**, a system monitoring result table **804**, and a node monitoring result table **805**. The server performance threshold table **801** and the storage performance threshold table **802** of the system monitoring information **320** are generated by the operations manager. The alert notification table **803**, the system monitoring result table **804**, and the node monitoring result table **805** are generated by automatic processing. The alert notification table **803** has the same number of records as alert notifications. To the system monitoring result table **804**, the same number of records as database servers are added every date and time of information collection. To the node monitoring result table **805**, the same number of records as worker nodes are added every date and time of information collection.

Each record of the server performance threshold table **801** is composed of fields of the threshold **806** of the CPU usage, a threshold **807** of the network usage, a threshold **808** of the disk transfer rate, a threshold **809** of the disk IOPS, a threshold **810** of the transfer rate of the device, a threshold **811** of the IOPS of the device. The threshold **806** of the CPU usage is a threshold used for abnormality sensing of the value of the CPU usage **710**. For example, the value of the CPU usage **710** is determined to be normal when falling within the range of the threshold, and is determined to be abnormal when being outside the range of the threshold. The same applies also to thresholds to be described below. The threshold **807** of the network usage is a threshold used for abnormality sensing of the value of the network usage **711**. The threshold **808** of the disk transfer rate is a threshold used for abnormality sensing of the value of the disk transfer rate **712**. The threshold **809** of the disk IOPS is a threshold used for abnormality sensing of the value of the disk IOPS **713**. The threshold **810** of the transfer rate of the device is a threshold used for abnormality sensing of the value of the transfer rate **732** of the device. The threshold **811** of the IOPS of the device is a threshold used for abnormality sensing of the value of the IOPS **733** of the device.

Each record of the storage performance threshold table **802** is composed of fields of a threshold **812** of the transfer rate of the port, a threshold **813** of the IOPS of the port, a threshold **814** of the transfer rate of the LDEV, and a threshold **815** of the IOPS of the LDEV. The threshold **812** of the transfer rate of the port is a threshold used for abnormality sensing of the value of the transfer rate **717** of the storage port. The threshold **813** of the IOPS of the port is a threshold used for abnormality sensing of the value of the IOPS **718** of the storage port. The threshold **814** of the transfer rate of the LDEV is a threshold used for abnormality sensing of the value of the transfer rate **737** of the LDEV. The threshold **815** of the IOPS of the LDEV is a threshold used for abnormality sensing of the value of the IOPS **738** of the LDEV.

Each record of the alert notification table **803** is composed of fields of date and time **816**, an alert type **817**, a system name **818**, a subject (RDB name or storage name) **819**, a metric type **820**, and an abnormal value **821**. The alert type **817** is the type of the alert. The subject **819** is an RDB name or storage name deemed as the subject of the alert. The metric type **820** and the abnormal value **821** show which metric is an abnormal value of what value.

Each record of the system monitoring result table **804** is composed of fields of date and time **822**, a system name **823**, a threshold determination result **824** of the CPU usage of the RDB server, a threshold determination result **825** of the network usage of the RDB server, a threshold determination result **826** of the disk IOPS of the RDB server, a threshold determination result **827** of the disk transfer rate of the RDB server, a threshold determination result **828** of the IOPS of the storage port, and a threshold determination result **829** of the transfer rate of the storage port.

The threshold determination result **824** of the CPU usage of the RDB server is the result of abnormality sensing by use of the threshold **806** of the CPU usage and the CPU usage **710**. The threshold determination result **825** of the network usage of the RDB server is the result of abnormality sensing by use of the threshold **807** of the network usage and the network usage **711**. The threshold determination result **826** of the disk IOPS of the RDB server is the result of abnormality sensing by use of the threshold **809** of the disk IOPS and the disk IOPS **713**. The threshold determination result **827** of the disk transfer rate of the RDB server is the result of abnormality sensing by use of the threshold **808** of the disk transfer rate and the disk transfer rate **712**. The threshold determination result **828** of the IOPS of the storage port is the result of abnormality sensing by use of the threshold **813** of the IOPS of the port and the IOPS **718** of the storage port. The threshold determination result **829** of the transfer rate of the storage port is the result of abnormality sensing by use of the threshold **812** of the transfer rate of the port and the transfer rate **717** of the storage port.

Each record of the node monitoring result table **805** is composed of fields of date and time **830**, a system name **831**, an RDB server name **832**, a node name **833**, a path **834** of the device, a threshold determination result **835** of the IOPS of the device, a threshold determination result **836** of the transfer rate of the device, a threshold determination result **837** of the transfer rate of the LDEV, and a threshold determination result **838** of the IOPS of the LDEV.

FIG. **8** is a diagram illustrating details of the scaling execution information **321**. The scaling execution information **321** is composed of an RDB scaling determination result table **901**, an RDB scale-up execution table **902**, an RDB scale-up result table **903**, a node movement determination result table **904**, an RDB scale-out execution table **905**, and a scale-out result table **906**. To the RDB scaling determination result table **901** of the scaling execution information **321**, the same number of records as database servers are added every time scaling determination is executed. To the RDB scale-up execution table **902** and the RDB scale-up result table **903**, as many records as the number of database servers about which it is determined that scale-up is necessary are added. In the node movement determination result table **904**, the same number of records as worker nodes are generated every time node movement determination is executed. To the RDB scale-out execution table **905** and the scale-out result table **906**, as many records as the number of database servers about which it is determined that scale-out is necessary are added.

Each record of the RDB scaling determination result table **901** is composed of fields of date and time **907**, a system name **908**, an RDB server name **909**, a server type **910**, and scaling necessity determination **911**. The scaling necessity determination **911** is any of scale-in, scale-out, scale-up, and scale-down.

Each record of the RDB scale-up execution table **902** is composed of fields of a task type **912**, a system name **913**, an RDB server name **914**, a server type **915**, a scale-up server type **916**, and a script program **917**. The server type **915** is the type of the present server before scale-up is executed. The scale-up server type **916** is the type of the server after the scale-up. The script program **917** is a script program used for scale-up processing.

Each record of the RDB scale-up result table **903** is composed of fields of a system name **918**, an RDB server name **919**, a before-scale-up server type **920**, and an after-scale-up server type **921**. The before-scale-up server type **920** and the after-scale-up server type **921** are the same kind as the server type **915** and the scale-up server type **916**, respectively.

Each record of the node movement determination result table **904** is composed of fields of date and time **922**, a system name **923**, an RDB server name **924**, a node name **925**, an LDEV name **926**, a storage port name **927**, scaling necessity determination **928**, and movement necessity determination **929**. The scaling necessity determination **928** is the necessity determination result of each of scale-in, scale-out, scale-up, and scale-down. The movement necessity determination **929** is the result of determination of whether or not movement to another virtual server in association with scale-up is necessary.

Each record of the RDB scale-out execution table **905** is composed of fields of a task type **930**, an RDB (coordinator) server name **931**, an RDB (worker) server name **932**, a server type **933**, the number of workers **934**, a worker node name (start number) **935**, and a script program **936**. The RDB (coordinator) server name **931** is the name of the server having a coordinator node. The RDB (worker) server name **932** is the name of the server having worker nodes. The number of workers **934** is the number of worker nodes that become the subject of scale-out processing. The worker node name (start number) **935** is the youngest number in identification numbers on the premise that the identification numbers of the worker nodes that become the subject of the scale-out processing are consecutive numbers. The script program **936** is a script program used for the scale-out processing.

Each record of the scale-out result table **906** is composed of fields of a before-movement RDB server name **937**, an after-movement RDB server name **938**, a before-movement node name **939**, an after-movement node name **940**, a before-movement storage port name **941**, an after-movement storage port name **942**, an LDEV ID **943**, and an LDEV name **944**.

FIG. **9** is a diagram illustrating details of the task execution information **322**. The task execution information **322** is composed of a task execution management table **1001**, a single server deployment table **1002**, a multi-server deployment table **1003**, an LDEV creation table **1004**, and an LDEV association table **1005**. The task execution information **322** is generated by the operations manager in some cases, and the system automatically registers the task execution information **322** on the basis of a scaling execution determination result in some cases. Related tables are created according to the task type of the task execution information **322**. For example, when the task type is "deployment," the single server deployment table **1002** or the multi-server deployment table **1003**, the LDEV creation table **1004**, and the LDEV association table **1005** are created.

Each record of the task execution management table **1001** is composed of fields of a task ID **1006**, a state **1007**, a task type **1008**, registration date and time **1009**, start date and time **1010**, ant end date and time **1011**. The task ID **1006** is an identifier to identify each task. The state **1007** is the state of each task and is, for example, "normal." The task type **1008** is the type of the task and is, for example, "system deployment," or "scale-out." The registration date and time **1009**, the start date and time **1010**, and the end date and time **1011** are the date and time when the task has been registered, started, and ended.

Each record of the single server deployment table **1002** is composed of fields of a task type **1012**, an RDB server name **1013**, a server type **1014**, the number of workers **1015**, and a script program **1016**. The number of workers **1015** is the number of worker nodes created at the time of deployment. In the present embodiment, the maximum number of worker nodes that can be created, for example, "64" or "256," is set. The script program **1016** is a script program for deploying a single server.

Each record of the multi-server deployment table **1003** is composed of fields of a task type **1017**, an RDB (coordinator) server name **1018**, an RDB (coordinator) server type **1019**, an RDB (worker) server name (start number) **1020**, an RDB (worker) server type **1021**, the number of workers **1022**, and a script program **1023**. The RDB (coordinator) server name **1018** is the name of the coordinator node. The RDB (coordinator) server type **1019** is the server type of the coordinator node. The RDB (worker) server name (start number) **1020** is the start number of numerals on the premise that consecutive numbers are used as the names of worker nodes. The RDB (worker) server type **1021** is the server type of the worker nodes.

Each record of the LDEV creation table **1004** is composed of fields of a task type **1024**, an LDEV ID (start number) **1025**, an LDEV name (start number) **1026**, an LDEV size **1027**, the number of volumes **1028**, and a script program **1029**. The LDEV size **1027** is the size of each logical volume configured in the storage apparatus **104**. The number of volumes **1028** is the total number of logical volumes configured in the storage apparatus **104**. As the number of volumes **1028**, a number equivalent to or larger than the number of workers **1015** is set. The script program **1029** is a script program that creates the logical volume in the storage apparatus **104**.

Each record of the LDEV association table **1005** is composed of fields of a task type **1030**, an RDB server name **1031**, a node name **1032**, an LDEV ID **1033**, an LDEV name **1034**, a storage port name **1035**, and a script program **1036**. In the LDEV association table **1005**, information that associates each LDEV generated in the storage apparatus **104** with a respective one of worker nodes generated in the virtual cloud **103** for business is stored. The script program **1036** is a script program that executes processing of associating the worker nodes with the LDEVs.

FIG. **10** is a diagram illustrating monitoring processing executed by the monitoring server **308**. However, the system table **601** and so forth are necessary to execute the monitoring processing flow and they are created in the process of automation processing to be described next. Thus, the monitoring processing to be described below becomes executable after execution of the system deployment program **315** to be described later in the automation processing has been completed.

In the monitoring processing, first, the in-system server information acquisition processing **405**, the in-server RDB configuration information acquisition processing **406**, and the related LDEV information acquisition processing **407** of the system configuration information collection program **310** are executed. The system configuration information collection program **310** reads in the system table **601**, the RDB server table **602**, the RDB node configuration table **603**, the node table **604**, and the LDEV table **605** and outputs the system configuration table **606**.

Next, the server performance data collection processing **408** and the storage performance data collection processing **409** of the system performance information collection program **311** are executed. The system performance information collection program **311** reads in the server performance table **701** and the storage port performance table **702** and outputs the system performance table **703**. Furthermore, the system performance information collection program **311** reads in the server device performance table **704** and the storage LDEV performance table **705** and outputs the node performance table **706**.

Next, the threshold determination processing **410** and the alert notification processing **411** of the system threshold determination program **312** are executed. In these kinds of processing, the system performance table **703**, the node performance table **706**, the server performance threshold table **801**, and the storage performance threshold table **802** are used and the system monitoring result table **804**, the node monitoring result table **805**, and the alert notification table **803** are output.

In the subsequent system monitoring result display program **313**, the system configuration information display processing **412** and the system performance information display processing **413** are executed. In these kinds of processing, information on the system configuration table **606**, the system performance table **703**, and the node performance table **706** are displayed.

FIG. **11** is a diagram illustrating the automation processing executed by the operations automation server **309**. In the automation processing, first, the task registration processing **501** and the task execution history display processing **503** of the task management program **314** are executed. In these kinds of processing, the task execution management table **1001** is read in.

Next, the VM deployment and RDB setting processing **504**, the storage volume allocation processing **505**, and the data load processing **506** of the system deployment program **315** are executed. The system deployment program **315** reads in the task execution management table **1001**, the single server deployment table **1002**, the multi-server deployment table **1003**, the LDEV creation table **1004**, and the LDEV association table **1005**. The system deployment program **315** outputs the system table **601**, the RDB server table **602**, the RDB node configuration table **603**, the node table **604**, and the LDEV table **605** and updates the task execution management table **1001**.

Next, the system threshold determination result acquisition processing **507** and the scaling necessity determination processing **508** of the scaling execution determination program **316** are executed. The scaling execution determination program **316** reads in the alert notification table **803**, the system monitoring result table **804**, and the node monitoring result table **805** and outputs the RDB scaling determination result table **901**, the node movement determination result table **904**, and the task execution management table **1001**.

Next, the scaling execution program **317** reads in the task execution management table **1001**, the RDB scaling determination result table **901**, and the node movement determination result table **904** and updates the task execution management table **1001**. The scaling execution program **317** executes any kind of processing on the basis of the description of the task execution management table **1001**. The scale-up processing **509** inputs the RDB scale-up execution table **902** and outputs the RDB scale-up result table **903**. The scale-down processing **510** inputs the RDB scale-up result table **903**. The scale-out processing **511** inputs the RDB scale-out execution table **905** and outputs the scale-out result table **906**. The scale-in processing **512** inputs the scale-out result table **906**.

FIG. **12** is a diagram illustrating processing executed by the system deployment program **315**. The VM deployment and RDB setting processing **504** of the system deployment program **315** first reads in the single server deployment table **1002** or the multi-server deployment table **1003**. The operations manager may indicate which table to read in by using means that is not illustrated in the diagram, or the operations manager may indirectly indicate which table to read in by inputting information to only either the single server deployment table **1002** or the multi-server deployment table **1003**.

In the VM deployment and RDB setting processing **504**, on the basis of the read-in single server deployment table **1002** or multi-server deployment table **1003**, an RDB server is deployed (S**1301**). Next, the system deployment program **315** installs a database management system (DBMS) (S**1302**), and executes initial setting of a cluster (S**1303**), and activates each node (S**1304**). The processing of the above-described S**1301** to S**1034** is implemented by executing the script program **1016** or the script program **1023**, for example.

Moreover, the system deployment program **315** outputs the system table **601**, the RDB server table **602**, the RDB node configuration table **603**, and the node table **604**. The output of these tables is implemented by executing the script program **1016** or the script program **1023**, for example. These script programs create the above-described tables with reference to information described in the single server deployment table **1002** or the multi-server deployment table **1003**. For example, information on the RDB server name **1013** in the single server deployment table **1002** is transcribed to the RDB server name **609** of the system table **601** and so forth.

In the storage volume allocation processing **505**, the system table **601**, the RDB server table **602**, the RDB node configuration table **603**, the node table **604**, the LDEV creation table **1004**, and the LDEV association table **1005** are read in, and the LDEV table **605** is output and the system table **601** is updated. In the storage volume allocation processing **505**, first an LDEV is created (S**1305**). Next, connection setting of the LDEV and the server (S**1306**) is executed. Then, setting that allows the LDEV to be used in a file format from the database server **107** (S**1307**) is executed and association of each node of the RDB server with the LDEV (S**1308**) is executed.

Specific processing for creating the LDEV in S**1305** is described in the script program **1029** of the LDEV creation table **1004**, for example. This script program **1029** operates with reference to other pieces of information included in the LDEV creation table **1004**. Specific processing for executing the processing of S**1306** to S**1308** is described in the script program **1036** of the LDEV association table **1005**, for example. This script program **1036** operates with reference to other pieces of information included in the LDEV association table **1005**.

FIG. **13** is a diagram illustrating details of the scaling necessity determination processing **508** executed by the scaling execution determination program **316**. In the scaling necessity determination processing **508**, the alert notification table **803**, the system monitoring result table **804**, and the node monitoring result table **805** are read in and the RDB scaling determination result table **901** and the node movement determination result table **904** are output. The scaling necessity determination processing **508** is composed of scaling necessity determination processing **1401** regarding each RDB server, scaling necessity determination processing **1402** regarding each node in the RDB server, and movement necessity determination processing **1403** regarding each RDB server node.

In the scaling necessity determination processing **1401** regarding each RDB server, the scaling execution determination program **316** first executes threshold excess determination of the device transfer rate (S**1401**). Specifically, the scaling execution determination program **316** compares the transfer rate **732** of the device in the server device performance table **704** with the threshold **810** of the transfer rate of the device in the server performance threshold table **801**. When determining that the device transfer rate exceeds the threshold, the scaling execution determination program **316** stores information showing that scale-out is necessary in the RDB scaling determination result table **901** (S**1405**). When determining that the device transfer rate does not exceed the threshold, the scaling execution determination program **316** proceeds to S**1406**.

In S**1406**, the scaling execution determination program **316** determines whether or not the CPU usage exceeds a threshold. Specifically, the scaling execution determination program **316** compares the CPU usage **710** in the server performance table **701** with the threshold **806** of the CPU usage in the server performance threshold table **801**. When determining that the CPU usage exceeds the threshold, the scaling execution determination program **316** stores information showing that scale-up is necessary in the RDB scaling determination result table **901** (S**1407**). When determining that the CPU usage does not exceed the threshold, the scaling execution determination program **316** stores information showing that scaling is unnecessary in the RDB scaling determination result table **901** (S**1408**).

In the scaling necessity determination processing **1402** regarding each node in the RDB server, the scaling execution determination program **316** first executes threshold excess determination of the LDEV transfer rate (S**1409**). Specifically, the scaling execution determination program **316** compares the transfer rate **737** of the LDEV in the storage LDEV performance table **705** with the threshold **814** of the transfer rate of the LDEV in the storage performance threshold table **802**. When determining that the LDEV transfer rate exceeds the threshold, the scaling execution determination program **316** stores, in the RDB scaling determination result table **901**, information showing that scale-out is necessary for the node associated with the LDEV of the determination subject in S**1409** (S**1410**). When determining that the LDEV transfer rate does not exceed the threshold, the scaling execution determination program **316** stores, in the RDB scaling determination result table **901**, information showing that scaling is not necessary for the node associated with the LDEV of the determination subject in S**1409** (S**1411**).

When it is determined that scale-out is necessary (S**1405**) in the scaling necessity determination processing **1401** regarding each RDB server, not only whether or not scale-out is necessary is determined regarding each node by the scaling necessity determination processing **1402** regarding each node in the RDB server, but whether or not scale-out is necessary may be determined on the basis of the magnitude of the IOPS of each node.

In the movement necessity determination processing **1403** regarding each RDB server node, the scaling execution determination program **316** refers to all node movement determination result tables **904** and determines whether or not the number of nodes for which scale-out is necessary is smaller than a predetermined threshold "n" (S**1412**). When determining that the number of nodes for which scale-out is necessary is smaller than "n," the scaling execution determination program **316** moves the nodes for which scale-out is necessary to newly-created nodes (S**1413**). When determining that the number of nodes for which scale-out is necessary is equal to or larger than "n," the scaling execution determination program **316** moves half of the nodes for which scale-out is necessary to newly-created nodes (S**1414**).

FIG. **14** is a diagram illustrating the scale-up processing **509** and the scale-down processing **510** executed by the scaling execution program **317**. In the scale-up processing **509**, the scaling execution program **317** reads in the RDB scaling determination result table **901** and the RDB scale-up execution table **902** and outputs the RDB scale-up result table **903**. Specifically, the scaling execution program **317** selects, as the processing subject, the database server about which it is described that scale-up is necessary in the scaling necessity determination **911** of the RDB scaling determination result table **901**, and executes processing of the following S**1501** and the subsequent steps.

The scaling execution program **317** first executes stop processing of the RDB server (S**1501**), and subsequently executes processing of making a change to the RDB server type described in the scale-up server type **916** in the RDB scale-up execution table **902** (S**1502**). Thereafter, the scaling execution program **317** executes activation processing of the RDB server (S**1503**) and outputs the RDB scale-up result table **903**. The pieces of information stored in the fields **918** to **921** of the RDB scale-up result table **903** are the same as the pieces of information stored in the fields **913** to **916** of the RDB scale-up execution table **902**.

The scaling execution program **317** executes the scale-down processing **510** when the processing load of the database server for which the scale-up processing has been executed becomes equal to or lower than a predetermined threshold. The scaling execution program **317** first stops the database server that becomes the subject of the scale-down processing (S**1504**). Next, the scaling execution program **317** changes the type of the server from the after-scale-up server type **921** of the RDB scale-up result table **903** to the before-scale-up server type **920** (S**1505**). Then, the scaling execution program **317** activates the server of the changed type (S**1506**).

FIG. **15** is a diagram illustrating the scale-out processing **511** and the scale-in processing **512** executed by the scaling execution program **317**. In the scale-out processing **511**, the scaling execution program **317** reads in the RDB scaling determination result table **901**, the node movement determination result table **904**, and the RDB scale-out execution table **905** and outputs the scale-out result table **906**. Specifically, the scaling execution program **317** selects, as the processing subject, the database server about which it is described that scale-out is necessary in the scaling necessity determination **911** of the RDB scaling determination result table **901**, and executes processing of the following S**1601** and the subsequent steps.

The scaling execution program **317** deploys a new virtual machine as an RDB server in the business system **105** (S**1601**). Next, the scaling execution program **317** refers to the LDEV name **926** of the node movement determination result table **904** and detects the LDEV that is the connection destination of the movement-target worker node (S**1602**). Then, the scaling execution program **317** connects the worker node of the new RDB deployed in S**1601** to the LDEV detected in S**1602** (S**1603**). Moreover, the scaling execution program **317** stops the movement-target worker node of the old RDB, i.e. the worker node about which the connection destination has been detected in S**1602** (S**1604**).

At last, the scaling execution program **317** activates the worker node of the new RDB, i.e. the worker node of the database server deployed in S**1601** (S**1605**). Then, the scaling execution program **317** describes necessary matters in the scale-out result table **906**.

The scaling execution program **317** starts the scale-in processing **512** when the transfer rate **732** of the device of the database server deployed in S**1601** is equal to or lower than a predetermined threshold and the transfer rate **737** of the LDEV becomes equal to or lower than a predetermined threshold regarding the LDEV associated with the worker node activated in S**1605**. In the scale-in processing **512**, first, the scaling execution program **317** stops the worker node in the RDB arising from scale-out (S**1606**), and refers to the LDEV ID **943** or the LDEV name **944** of the scale-out result table **906** and detects the connection-destination LDEV of the worker node in the RDB arising from the scale-out. (S**1607**)

Next, the scaling execution program **317** identifies the before-movement database server and worker node by referring to the before-movement RDB server name **937** and the before-movement node name **939** and reattaches the detected LDEV to the stopped worker node in the before-scale-out RDB (S**1608**). Then, the scaling execution program **317** activates the worker node stopped in the scale-out processing (S**1609**) and deletes the worker node of the database server arising from the scale-out from the coordinator node (S**1610**). At last, the scaling execution program **317** deletes the worker node in the database server arising from the scale-out (S**1611**).

FIG. **16** and FIG. **17** are screen display examples of an execution result illustrating change in the system performance in association with scale-out processing and scale-in processing. In the example illustrated in FIG. **16** and FIG. **17**, deployment of the system is started at a clock time t**0** and a job involving a high load is started at a clock time t**1**. Then, scale-out is started at a clock time t**2** and the above-described job ends at a clock time t**3**. Then, scale-in is started at a clock time t**4**.

In FIG. **16**, a task execution history **1701**, a before-scale-out system configuration **1702**, and an after-scale-out system configuration **1703** are illustrated. In the task execution history **1701**, the type of the task and the clock time of start of the task are included. Specifically, the deployment of the system at the clock time t**0**, the start of the scale-out at the clock time t**2**, and the start of the scale-in at the clock time t**4** are described. In the before-scale-out system configuration **1702**, the system configuration in the state before the scale-out is started, i.e. before the clock time t**2**, is shown. In the after-scale-out system configuration **1703**, the system configuration after the scale-out and before the scale-in is shown.

When the before-scale-out system configuration **1702** and the after-scale-out system configuration **1703** are compared with each other, it turns out that "worker2" moves from "VM01" to "VM02" and the name thereof is changed to "worker3" and "LDEV2" moves from "/mnt/disk2" to "/mnt/disk3." Furthermore, the storage port is changed from only "CL1-A" to "CL1-A" and "CL2-A." Although only two states of the before-scale-out system configuration **1702** and the after-scale-out system configuration **1703** are illustrated in FIG. **16**, a system configuration of a state specified by a user may be shown and all system configurations may be shown.

FIG. **17** is a diagram illustrating time-series change in the system performance corresponding to FIG. **16**. Specifically, in FIG. **17**, time-series change graphs of the server CPU usage, the server device transfer rate, and the storage port transfer rate are illustrated. In all of the three graphs illustrated in FIG. **17**, the time elapses in the right direction of the diagram and the clock times correspond with each other in the vertical direction. A dashed line of the horizontal direction illustrated in each of the three graphs of FIG. **17** shows a respective one of thresholds. Specifically, the thresholds are the threshold **806** of the CPU usage, the threshold **808** of the disk transfer rate, and the threshold **812** of the transfer rate of the port.

When the job involving a high load is started at the clock time t**1**, thereafter the CPU usage and the server device transfer rate of "VM1" and the storage port transfer rate of "CL1-A" rise. Then, the server device transfer rate of "VM1" exceeds the threshold. Therefore, an affirmative determination is made in S**1404** in FIG. **13** and the scale-out is executed at the clock time t**2**. When "worker3" starts operation in "VM2" newly deployed by scale-out processing, the server device transfer rate of "VM1" decreases. Thereafter, when the job ends at the clock time t**3**, the server device transfer rate of "VM2" decreases and scale-in processing is executed at the clock time t**4**, so that a return to the state before the execution of the scale-out processing is made.

(Related Art)

FIG. **18** is a diagram illustrating one example of scale-out processing in a configuration of a related art (hereinafter, "related-art configuration") **200**Z. In the related-art configuration **200**Z, the scale-out processing is executed as follows when the I/O transfer rate of a virtual NIC **115**Z of a database server **107**Z exceeds a threshold. Specifically, a database server **116**Z is newly created and a third worker node **117**Z is created in the database server **116**Z and is added to a coordinator node **109**Z. Moreover, in storage apparatus **104**Z, a third LDEV **201**Z is newly created and is coupled to the third worker node **117**Z by plural connections via a storage port **118**Z.

Thereafter, data stored in a first LDEV **112**Z that stores data of a first worker node **110**Z in the database server **107**Z and a second LDEV **114**Z that stores data of a second worker node **1112** is redistributed to the third LDEV **201**Z that stores data of the third worker node **117**Z. When this redistribution of the data is executed between a public cloud **1012** and a near cloud **102**Z, a long time is required for data transfer. For this reason, a long time is required until the third worker node **117**Z starts processing and it is impossible to respond to a sudden increase in the processing load.

According to the above-described first embodiment, the following operation and effects are obtained.

(1) The scaling system **100** is configured to range over the public cloud **101** in which the database server **107** in which the first worker node **110** and the second worker node **111** operate is set and the near cloud **102** in which the storage apparatus **104** connected to the public cloud **101** by a network is set. The storage apparatus **104** includes the first

storage port **113** and the second storage port **118**. The storage apparatus **104** includes the first volume **112** accessed by the first worker node **110** and the second volume **114** accessed by the second worker node **111**. The VM **116** is set in the public cloud **101**. The second worker node **111** is moved to the VM **116** and is caused to operate as the third worker node **117** and the second volume **114** is caused to communicate with the third worker node **117** through the second storage port **118** if the transfer rate of the database server **107** exceeds a predetermined threshold (FIG. **13**, S**1404**: True) or if the transfer rate of the first storage port **113** exceeds a predetermined threshold (FIG. **13**, S**1409**: True) when the first volume **112** and the second volume **114** are executing communication with the database server **107** through the first storage port **113**. Thus, reconfiguration of the logical volume in association with the scale-out processing is unnecessary. Therefore, the problem as in the related art illustrated in FIG. **18** does not occur and it is possible to respond to a sudden increase in the processing load.

(2) The database server **107** and the VM **116** are virtual machines. Therefore, scale-up and scale-down can be easily executed according to the calculation load. Furthermore, for the virtual machine, the upper limit of the transfer rate is set lower than that in bare-metal apparatus in many cases. Therefore, the necessity of application of the present invention is high and effects of the present invention are readily obtained.

(3) The scaling system **100** is constructed to range over the public cloud **101** allowed to be used by a large number of unspecified persons and the near cloud **102** allowed to be used by only specific persons. For this reason, flexible change in calculation resources is easy in the public cloud **101** and using the near cloud **102** allows data to be safely managed.

(4) The maximum number of worker nodes allowed to be created are created in the database server **107** and the storage apparatus **104** has at least the same number of volumes as the worker nodes. Therefore, for the calculation system S, scale-out can be repeated plural times according to need.

(5) Each of the first worker node **110** and the second worker node **111** is a node that implements a distributed database. Therefore, for the calculation system S, scale-out of a distributed database can be rapidly implemented.

(6) Half of the worker nodes included in the database server **107** are moved to the VM **116** when the number of worker nodes about which the disk IOPS is equal to or higher than a predetermined threshold or the transfer rate of the corresponding logical volume is equal to or higher than a predetermined threshold is equal to or larger than a predetermined threshold "n" (FIG. **13**, S**1412**: False, S**1414**). The worker nodes about which the disk IOPS is equal to or higher than the predetermined threshold or the transfer rate of the corresponding logical volume is equal to or higher than the predetermined threshold are moved to the VM **116** when the number of worker nodes about which the disk IOPS is equal to or higher than the predetermined threshold or the transfer rate of the corresponding logical volume is equal to or higher than the predetermined threshold is smaller than the predetermined threshold "n" (FIG. **13**, S**1412**: True, S**1413**).

(7) The scaling system **100** executes scale-in processing to cause the third worker node to operate as the second worker node in the first calculating apparatus and cause the second volume to communicate with the second worker node through the first network port. Therefore, the scale-in processing can be executed after the scale-out processing.

(8) Calculation resources of the first calculating apparatus are scaled up when the CPU usage of the first calculating apparatus exceeds a predetermined threshold and the disk IOPS of the first calculating apparatus is equal to or lower than the predetermined threshold. Therefore, in the calculation system S, the scale-up is possible according to need.

(9) The scaling system **100** includes the system monitoring result display program **313** that displays time-series change in the system performance of the scaling system **100** and the configuration of the scaling system **100** as illustrated in FIG. **16** and FIG. **17**. Therefore, the state of the system can be presented to the user in an easy-to-understand manner.

## Modification Example 1

In the above-described embodiment, the virtual machines in the public cloud **101** included in the scaling system **100** are only two, the database server **107** and the VM **116**. However, three or more virtual machines may exist in the public cloud **101** included in the scaling system **100**. Furthermore, since the maximum number that can be set is desirable as the number of worker nodes as described above, at most the same number of virtual machines as the worker nodes can be used according to increase in the amount of transfer.

## Modification Example 2

In the above-described embodiment, the storage apparatus **104** may be placed in not the near cloud **102** but an on-premise environment. Furthermore, the calculation system S may include plural pieces of storage apparatus **104**.

## Modification Example 3

In the above-described embodiment, the scaling system **100** has the system configuration information **318**, the system performance information **319**, the system monitoring information **320**, the scaling execution information **321**, and the task execution information **322** as illustrated in FIG. **2** and FIG. **5** to FIG. **9**. However, all pieces of information of the system configuration information **318**, the system performance information **319**, the system monitoring information **320**, the scaling execution information **321**, and the task execution information **322** are not necessary and it suffices that the scaling system **100** has information that allows execution of the processing explained in the embodiment.

## Modification Example 4

In the above-described embodiment, the calculation system S implements a distributed relational database. However, the database implemented by the calculation system S is not limited to the relational database. Furthermore, the calculation system S may implement an element other than the database.

In the above-described embodiment and modification examples, the configurations of functional blocks are merely one example. Several functional configurations illustrated as separate functional blocks may be integrally configured and a configuration represented by one functional block diagram may be divided into two or more functions. Furthermore, a configuration may be employed in which part of functions that the respective functional blocks have is possessed by another functional block.

The above-described embodiment and modification examples may be combined with each other. Although the various embodiment and modification examples have been explained in the above description, the present invention is not limited to the contents of them. Other aspects that are conceivable in the range of technical ideas of the present invention are also included in the range of the present invention.

What is claimed is:

1. A scaling system for improving resource utilization and security in a hybrid cloud computing environment, the scaling system comprising:

a first base that includes a first calculating apparatus, and a first worker node and a second worker node operate within the first calculating apparatus;

a second base that includes storage apparatus connected to the first base by a network, wherein the storage apparatus includes a first network port, a second network port, a first volume accessed by the first worker node via the first network port, and a second volume accessed by the second worker node via the first network port,

wherein access to the second base is restricted to fewer users than the first base; and

a monitoring server that is communicatively coupled to the first base and the second base, wherein the monitoring server is configured to:

determine that a transfer rate of the first calculating apparatus or a transfer rate of the first network port exceeds a predetermined threshold when the first volume and the second volume are executing communication with the first calculating apparatus through the first network port, and

in response to determining that the transfer rate of the first calculating apparatus or the transfer rate of the first network port exceeds the predetermined threshold when the first volume and the second volume are executing communication with the first calculating apparatus through the first network port:

the first volume remains coupled to the first worker node via the first network port,

add a second calculating apparatus to the first base, move the second worker node to the second calculating apparatus, and

once the second worker node is moved, cause the second worker node to operate as a third worker node, and configure the second volume to couple with the third worker node via the second network port.

2. The scaling system according to claim **1**, wherein

the first calculating apparatus and the second calculating apparatus are virtual machines.

3. The scaling system according to claim **1**, wherein

the first base is a public cloud allowed to be used by a plurality of unspecified persons, and

the second base is a near cloud or an on premise environment allowed to be used by only specific persons.

4. The scaling system according to claim **1**, wherein

a maximum number of worker nodes allowed to be created are created in the first calculating apparatus, and

the storage apparatus has at least a same number of volumes as the worker nodes.

5. The scaling system according to claim **1**, wherein

each of the first worker node and the second worker node is a node that implements a distributed database.



6. The scaling system according to claim **4**, wherein

on a condition that the first volume and the second volume are executing communication with the first calculating apparatus through the first network port and the transfer rate of the first calculating apparatus or the transfer rate of the first network port exceeds the predetermined threshold;

half of the worker nodes included in the first calculating apparatus are moved to the second calculating apparatus when a number of the worker nodes included in a disk input/output operation per second (IOPS) is equal to or larger than a predetermined threshold number of nodes, and

the worker nodes included in the disk IOPS are moved to the second calculating apparatus when the number of worker nodes included in the one disk IOPS is smaller than the predetermined number of nodes, and

the disk input/output operation per second (IOPS) is equal to or higher than a predetermined threshold or a transfer rate of a corresponding volume is equal to or higher than a predetermined threshold.

7. The scaling system according to claim **1**, wherein

the scaling system further executes scale-in processing to cause the third worker node to operate as the second worker node in the first calculating apparatus and cause the second volume to communicate with the second worker node through the first network port.

8. The scaling system according to claim **1**, wherein

calculation resources of the first calculating apparatus are scaled up when CPU usage of the first calculating apparatus exceeds a predetermined threshold and a disk input/output operation per second (IOPS) of the first calculating apparatus is equal to or lower than a predetermined threshold.

9. The scaling system according to claim **1**, wherein

the scaling system further includes a system monitoring result display section that displays time-series change in system performance of the scaling system and a configuration of the scaling system.

10. A calculation method executed by a computer included in a scaling system, for improving resource utilization and security in a hybrid cloud computing environment, is configured to range over a first base that includes a first calculating apparatus, and a first worker node and a second worker node operate within the first calculating apparatus and a second base that includes a storage apparatus connected to the first base by a network,

wherein the storage apparatus includes a first network port, a second network port, a first volume accessed by the first worker node via the first network port, and a second volume accessed by the second worker node via the first network port, and

wherein access to the second base is restricted to fewer users than the first base,

the calculation method comprising:

determining that a transfer rate of the first calculating apparatus or a transfer rate of the first network port exceeds a predetermined threshold when the first volume and the second volume are executing communication with the first calculating apparatus through the first network port, and

in response to determining that the transfer rate of the first calculating apparatus or the transfer rate of the first network port exceeds the predetermined threshold when the first volume and the second volume are executing communication with the first calculating apparatus through the first network port:

the first volume remains coupled to the first worker node via the first network port,

adding a second calculating apparatus to the first base,

moving the second worker node to the second calculating apparatus, and

once the second worker node is moved, causing the second worker node to operate as a third worker node, and configuring the second volume communicates with the third worker node through the second network port.

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